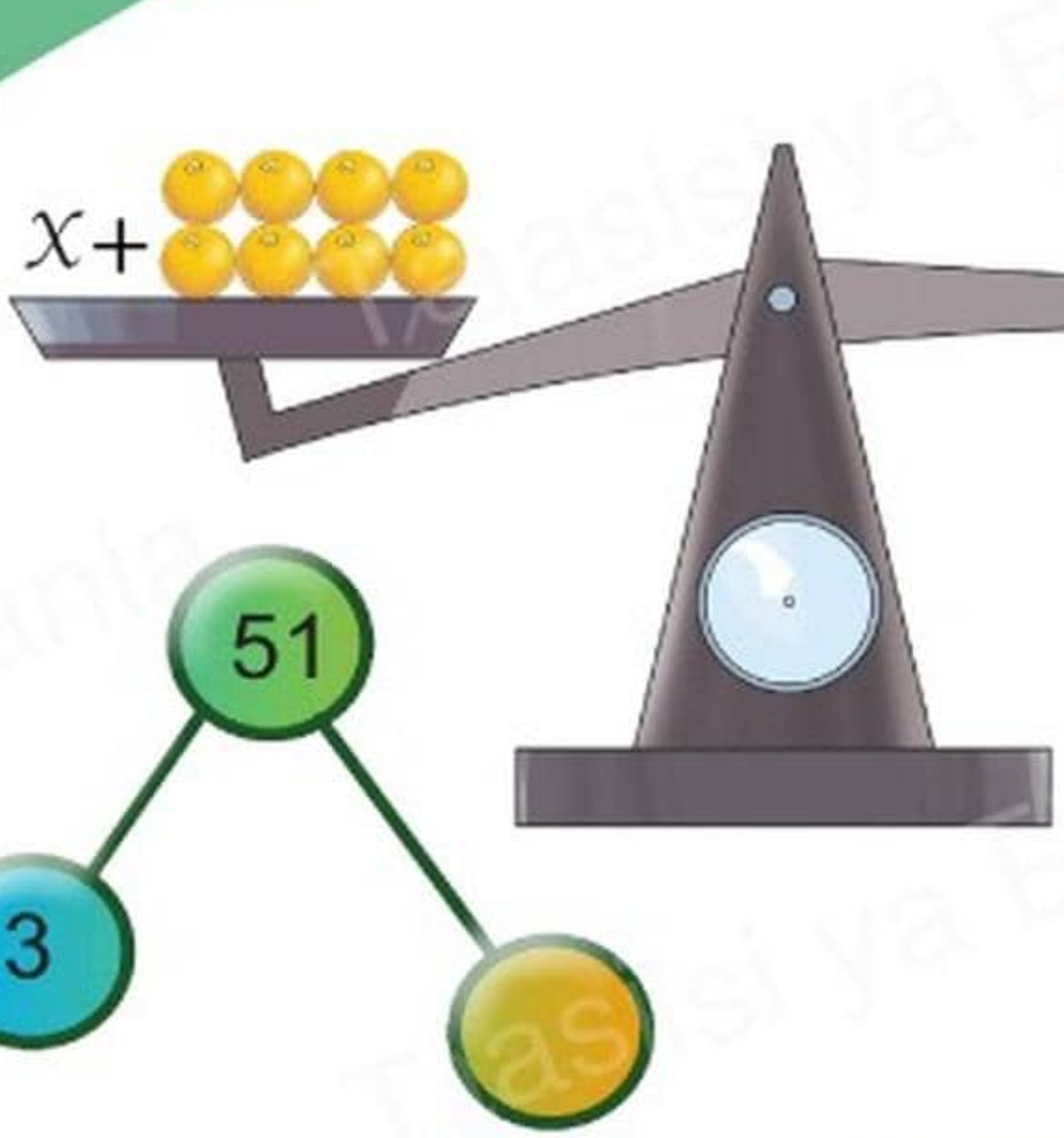
Mathematics

Pupil's Book

Standard Five



Tanzania Institute of Education

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Mathematics

Pupil's Book

Standard Five

THE UNITED REPUBLIC OF TANZANIA

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Dr Lyabwene M. Mtahabwa

Commissioner for Education

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Dr Aneth A. Komba
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Introduction

This textbook is prepared for a Standard Five Pupil in Tanzania Mainland. Mathematics Pupil's Book for Standard Five is written per the 2023 Mathematics Syllabus for Primary School Education Standard III – VI issued by the Ministry of Education, Science and Technology. This is the second edition after the improvement of the Standard Five Textbook published in 2019 as based on the 2016 Mathematics Syllabus for Primary School Education Standard III – VII.

This book consists of seven chapters: Measurements of length, mass and volume; Circumference and area of a circle; Types, factors and multiples of numbers; Fractions; Decimals; Algebra; and Financial mathematics. The competencies developed will enable you to acquire basic mathematical skills and their applications, logical thinking and solving real-life problems.

Content of the chapters are presented interms of texts, drawings, pictures, activities and exercises. You are advised to engage yourself and with friends in all activities and exercises in this book and other works provided with your teacher to develop the intended competencies.

Additional learning resources are available in the TIE-e-Library at *https://ol.tie.go.tz* or *ol.tie.go.tz*

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Chapter One

Measurements of length, mass and volume

Introduction

In this chapter, you will learn about multiplication and division of metric units of length, mass and volume. The competencies developed will enable you to perform various measurement in constructions, business, tailoring, cooking, hospitals, agriculture and other uses in daily life.

Think

Life without measurements

Multiplication of metric units

Measurements of length

In order to multiply the metric units of length, it is important to know the relationship between the metric units of length from millimetres to kilometres.

Activity 1 **Measuring lengths of various things**

Steps

1. Measure and record the length of the top of your desk or table in centimetres.
2. Convert the length you obtained in Step 1 into milimetres and decimetres.
3. Measure and record the length of your Standard Five Mathematics textbook in centimetres.

4. If there are 45 pupils in your class and each pupil has a textbook, what is the total length of the textbooks in metres and centimetres?
5. Use a tape measure to measure the length of your classroom. Record the obtained length in metres.
6. Convert the obtained length in Step 5 into centimetres and millimetres. What will be the total length of 15 classrooms of such kind?

Example 1

A carpenter made a table with a length of 3 m 30 cm and 5 mm. What is the total length of 12 tables of the same size in metres, centimetres and millimetres?

Steps	Solution
1. Arrange the metric units in columns with the smallest unit on the right and the largest unit on the left	m cm mm 3 30 5 × 12
2. Start multiplying the column of 5 mm by 12 to get 60 mm. Remember, $1\text{ cm} = 10\text{ mm}$. Thus, 60 mm is equal to 6 cm 3. Write 0 in the mm column 4. Multiply the centimetres column: $30\text{ cm} \times 12 = 360\text{ cm}$ 5. Add 360 cm to 6 cm obtained from the mm column. The total is 366 cm. Remember, $1\text{ m} = 100\text{ cm}$. Thus, 366 cm is equal to 3 m and 66 cm 6. Write 66 in the cm column	m cm mm 3 30 5 × 12 39 66 0

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7. Multiply the metres column: $3\,\text{m} \times 12 = 36\,\text{m}$ then add 3 m obtained from the cm column: $36\,\text{m} + 3\,\text{m} = 39\,\text{m}$ 8. Write 39 in the m column	

Therefore, 12 tables will have a total length of $39\ \text{m}$ and $66\ \text{cm}$.

Example 2

Multiply 3 km 604 m 54 cm by 4

Steps	Solution
1. Arrange the units in columns, with the smallest unit on the right and the largest unit on the left	km m cm 3 604 54 × 4
2. Multiply from right to left or from the smallest to the largest unit 3. Multiply by considering the relationship between units of length, for example: $1\ \text{km} = 1000\ \text{m}$, $1\ \text{m} = 100\ \text{cm}$ 4. Multiply 54 cm by 4 to get 216 cm. Remember, $1\ \text{m} = 100\ \text{cm}$; therefore, 216 cm is equal to 2 m and 16 cm 5. Write 16 in the cm column 6. Multiply 604 m by 4 to get 2416 m. Add 2 m that remained in step 4 to 2416 m. That is, $2416\ \text{m} + 2\ \text{m} = 2418\ \text{m}$	km m cm 3 604 54 × 4 14 418 16

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7. Remember, $1\ \text{km} = 1000\ \text{m}$; therefore, 2418 m is equal to $2\ \text{km}$ and $418\ \text{m}$ 8. Write 418 in the m column 9. Multiply 3 km by 4 to get 12 km, then add 2 km that remained in step 7 to 12 km: $12\ \text{km} + 2\ \text{km} = 14\ \text{km}$ 10. Write 14 in the km column	

Therefore, the answer is $14\ \text{km}\ 418\ \text{m}\ 16\ \text{cm}$.

Example 3

Multiply 8 km 704 dam 513 dm by 5

Steps	Solution
1. Arrange the units in columns, with the smallest unit on the right and the largest unit on the left	km dam dm 8 704 513 × 5
2. Multiply 513 dm by 5 to get 2565 dm. Remember, $1\ \text{dam} = 100\ \text{dm}$; therefore, 2565 dm is equal to 25 dam and 65 dm 3. Write 65 in the dm column 4. Multiply 704 dam by 5 to get 3520 dam. Add 3520 dam and 25 dam, which is obtained from step 2; $3520\ \text{dam} + 25\ \text{dam} = 3545\ \text{dam}$ 5. Remember, $1\ \text{km} = 100\ \text{dam}$; therefore 3545 dam is equal to 35 km and 45 dam	km dam dm 8 704 513 × 5 75 45 65

6. Write 45 in the dam column
7. Multiply 8 km by 5 to get 40 km, then add 40 km and 35 km, which is obtained from step 5; $40\ \text{km} + 35\ \text{km} = 75\ \text{km}$
8. Write 75 in the km column

Therefore, the answer is $75\ \text{km}\ 45\ \text{dam}\ 65\ \text{dm}$.

Example 4

m	cm	mm
7	21	4
×		8

Solution

m	cm	mm
7	21	4
×		8
57	71	2

Therefore, the answer is $57\ \text{m}\ 71\ \text{cm}\ 2\ \text{mm}$.

Exercise 1

1.

km	m	cm
1	510	21
×		9

2.

km	m	cm
10	133	44
×		14

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3.

m	cm	mm
17	72	9
×		2

4.

km	dam	dm
15	95	62
×		8

5.

km	m	cm
8	2	9
×		5

6.

dm	cm	mm
7	1	4
×		8

7.

km	dam	dm
27	42	31
×		18

8.

km	m	cm
38	31	4
×		29

9.

km	m	cm
4	18	42
×		10

10.

km	m	cm
57	82	92
×		5

11. In Makanya town there are two roads, the first road is 257 kilometres and 625 metres long and the second road is 5 times longer than the first road. What is the length of the second road?

12. The bus travelled from Dodoma region to Morogoro region with an average speed of 75 kilometres per hour. How long did the bus travel after 4 hours?

13. Nine books of the same type have the same width. Find the total width of those books in cm and mm, if the width of each book is 3 cm and 8 mm.

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Measurements of mass

In order to multiply units of mass, you are required to understand the relationship between metric units of mass from milligrams to tonnes.

Activity 2 Measuring of different objects

Steps

1. Use a beam balance to measure various objects from your environment like empty bottle, bottle filled with water, a paper, a pen, exercise book, stones, sand, pupil bag and a box of chalks.
2. Record the mass of the objects in kilograms and grams.
3. Convert the recorded masses in Step 2 into other three units of mass.
4. Among the recorded metric units of mass in Step 3 which metric unit is the largest? Why?

Example 1

Multiply 3 t 964 kg 300 g by 5

Steps	Solution
1. Arrange the units in columns with the smallest unit on the right and the largest unit on the left	t kg g 3 964 300 × 5
2. Multiply from right to left or from the smallest to the largest unit 3. Remember; 1 tonne = 1000 kg, 1 kg = 1000 g	

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Steps	Solution
4. Multiply 300 g by 5 to get 1500 g. Remember, $1\ \text{kg} = 1000\ \text{g}$. Therefore, 1500 g is equal to $1\ \text{kg}$ and $500\ \text{g}$ 5. Write 500 in the g column 6. Multiply 964 kg by 5 to get 4820 kg, then add 1 kg to get 4821 kg. Remember, $1\ \text{tonne} = 1000\ \text{kg}$. Therefore, 4821 kg is equal to 4 tonnes and 821 kg 7. Write 821 in the kg column 8. Multiply 3 tonnes by 5 to get 15 tonnes, then add 4 tonnes which is obtained in step 6; $15\ \text{tonnes} + 4\ \text{tonnes} = 19\ \text{tonnes}$ 9. Write 19 in the tonnes column	t kg g 3 964 300 × 5 19 821 500

Therefore, the answer is $19\ \text{tonnes}\ 821\ \text{kg}\ \ 500\ \text{g}$.

Example 2

Multiply 5 dag 14 dg 15 mg by 8

Steps	Solution
1. Arrange the units in columns, with the smallest unit on the right and the largest unit on the left	dag dg mg 5 14 15 × 8
2. Start multiplying small units to larger units. Multiply 15 mg by 8 to get 120 mg. Remember, $1\ \text{dg} = 100\ \text{mg}$. Thus, 120 mg is equal to 1 dg and 20 mg	

3. Write 20 in the mg column
4. Multiply 14 dg by 8 to get 112 dg then add 1 dg to get 113 dg. Remember, $1 \text{ dag} = 100 \text{ dg}$. Therefore, 113 dg is equal to 1 dag and 13 dg
5. Write 13 in the dg column
6. Multiply 5 dag by 8 to get 40 dag, then add 1 dag which is obtained in step 4; $40 \text{ dag} + 1 \text{ dag} = 41 \text{ dag}$
7. Write 41 in the dag column

dag	dg	mg
5	14	15
×		8
41	13	20

Therefore, the answer is $41 \text{ dag } 13 \text{ dg } 20 \text{ mg}$.

Example 3

A goat gave birth to three kids at once. Each kid weighed an average of 3 kilograms and 250 grams. What was the total mass of all the kids?

Solution

Arrange the units of mass in columns then follow steps on multiplication of metric units.

kg	g
3	250
×	3
9	750

Therefore, the total mass of all kids is 9 kg and 750 g.

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Exercise 2

1.

t	kg	dag
30	104	34
×		5

2.

dag	dg	mg
8	23	21
×		14

3.

dag	dg	mg
9	41	35
×		11

4.

dag	dg	mg
12	51	15
×		6

5.

t	kg	dag
26	73	62
×		7

6.

dag	dg	mg
11	62	25
×		2

7.

dag	dg	mg
35	53	32
×		16

8.

t	kg	dag
64	48	60
×		10

9.

t	kg	dag
45	81	31
×		9

10.

dag	dg	mg
65	44	57
×		15

11. A can of oil weighs 16 kg and 130 g. What is the mass of a crate with 22 such cans if the mass of the empty crate is ignored?

12. A goat weighs 20 kg and 15 g. What is the total mass in kilograms and grams of 16 goats of the same mass?

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13. Forty-five cattle hides had the same mass. Find the total mass of all the hides in kg and g if each hide weighed 10 kg and 24 g.

Measurements of volume

When multiplying metric units of volume, start multiplying the smallest measurement to the largest measurement considering the relationship between metric units of volume from millilitres to litres.

Activity 3 — Measuring the volume of water by using bottles

Collect three bottles whose capacities are 500 mL, 1000 mL and 1500 mL.

Steps

1. Use a 500 mL bottle to fill a 1000 mL bottle.
2. How many 500 mL bottles will be used to fill a 1000 mL bottle?
3. Use 500 mL bottle to fill a 1500 mL bottle.
4. How many 500 mL bottles will be used to fill 1500 mL bottle.
5. If you have 10 bottles with 1.5 L and 500 mL capacities, what is their total volume? Give your answer in litres and millilitres.

Example 1

Multiply 23 L 200 mL by 7

Steps

1. Arrange the units in columns with the smallest unit on the right and the largest unit on the left

L	mL
23	200
×	7

2. Consider the relationship between units of volume; $1\ \text{L} = 1000\ \text{mL}$

3. Multiply 200 mL by 7 to get 1,400 mL. Remember, $1\ \text{L} = 1000\ \text{mL}$. Therefore, 1400 mL is equal to 1 L and 400 mL 4. Write 400 in the mL column 5. Multiply 23 L by 7 to get 161 L, then add 161 litres and 1 litre, which is obtained in step 3; $161\ \text{L} + 1\ \text{L} = 162\ \text{L}$ 6. Write 162 in the L column	L mL 23 200 × 7 162 400

Therefore, the answer is $162\ \text{L}\ \ 400\ \text{mL}$.

Example 2

Multiply 45 L 612 mL by 9

Steps	Solution
1. Arrange the units in columns, with the smallest unit on the right and the largest unit on the left	L mL 45 612 × 9
2. Multiply 612 mL by 9 to get 5508 mL. Remember, $1\ \text{L} = 1000\ \text{mL}$. Therefore, 5508 mL is equal to 5 L and 508 mL 3. Write 508 in the mL column 4. Multiply 45 L by 9 to get 405 L. Add 405 L and 5 L which is obtained in step 3; $405\ \text{L} + 5\ \text{L} = 410\ \text{L}$ 5. Write 410 in the L column	L mL 45 612 × 9 410 508

Therefore, the answer is $410\ \text{L}\ \ 508\ \text{mL}$.

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Example 3

Multiply 35 L 314 mL by 15

Solution

	L	mL
	35	314
×		15
	529	710

Therefore, the answer is 529 L 710 mL.

Exercise 3

1.

	L	mL
	50	600
×		2

2.

	L	mL
	81	12
×		10

3.

	L	mL
	12	707
×		14

4.

	L	mL
	21	421
×		5

5.

	L	mL
	104	223
×		10

6.

	L	mL
	3	6
×		200

7.

	L	mL
	82	22
×		5

8.

	L	mL
	64	47
×		15

9.

	L	mL
	61	161
×		14

10.

	L	mL
	38	78
×		32

11.

	L	mL
	62	416
×		6

12.

	L	mL
	54	100
×		10

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13.

L	mL
23	100
×	6

14.

L	mL
23	5
×	90

15. In a group of 60 people, each person used 25 litres and 750 millilitres of water. How many litres were used by all people?

Division of metric units

The steps for dividing metric units are not different from those of multiplication. The important thing to remember is the relationship between the metric units. The only difference is that when dividing metric units, you start from the largest to the smallest unit.

Example 1

A piece of fabric whose length is 30 m and 60 cm was divided into 6 equal pieces. What was the length of each piece?

Steps	Solution
1. Arrange units of length in columns, then insert the division symbol	m cm 6) 30 60
2. Divide by considering the relationship of the units of length. For example, 1 m = 100 cm 3. Divide 30 m by 6, to get 5 m. Write 5 in the m column 4. Divide 60 cm by 6 to get 10 cm 5. Write 10 in the cm column	m cm 5 10 6) 30 60

Therefore, the answer is 5 m 10 cm.

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Example 2

Divide 3 m 84 cm 5 mm by 5

Steps	Solution
1. Arrange units of length in columns, then insert the division symbol	$5\overline{)\begin{array}{lll} \text{m} & \text{cm} & \text{mm} \\ 3 & 84 & 5 \end{array}}$
2. Divide by considering the relationship of the units of length. For example, $1\text{ m} = 100\text{ cm}$, $1\text{ cm} = 10\text{ mm}$	
3. Divide 3 m by 5. It is not sufficient to get a whole number. Thus, write 0 in the m column	
4. Remember, $1\text{ m} = 100\text{ cm}$; convert 3 m into cm by multiplying 3 by 100 to get 300 cm	
5. Add 300 cm to 84 cm to get 384 cm	
6. Divide 384 cm by 5 to get 76 cm and the remainder is 4 cm	
7. Write 76 in the cm column	$5\overline{)\begin{array}{lll} \text{m} & \text{cm} & \text{mm} \\ 0 & 76 & 9 \\ 3 & 84 & 5 \end{array}}$
8. Remember, $1\text{ cm} = 10\text{ mm}$ thus, convert 4 cm that remained in step 6 into mm by multiplying 4 cm by 10 to get 40 mm	
9. Add 40 mm and 5 mm to get 45 mm	
10. Divide 45 mm by 5 to get 9 mm	
11. Write 9 in the mm column	

Therefore, the answer is $76\text{ cm }9\text{ mm}$.

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Example 3

5000 L and 240 mL of oil are to be stored in 8 barrels of the same capacity. How many litres of oil will each barrel hold?

Solution

To know the capacity of each barrel, divide the given volume by the number of barrels.

Arrange units of volume in columns, then follow the steps of dividing the units of volume.

$$\begin{array}{r} \text{L} \quad \text{mL} \\ 8\overline{)\,5000 \quad 240} \end{array} = \begin{array}{l} \text{L} \quad \text{mL} \\ 625 \quad 30 \end{array}$$

Therefore, each barrel will hold 625 L and 30 mL of oil.

Example 4

Divide 48 kg 564 g by 6

Steps	Solution
1. Consider the relationships between metric units of mass. For example: $1\text{ kg} = 1000\text{ g}$	$\begin{array}{r} \text{kg} \quad \text{g} \\ 6\overline{)\,48 \quad 564} \end{array}$
2. Divide 48 kg by 6, to get 8 kg 3. Write 8 in the kg column 4. Divide 564 g by 6 to get 94 g 5. Write 94 in the g column	$\begin{array}{r} \text{kg} \quad \text{g} \\ 8 \quad 94 \\ 6\overline{)\,48 \quad 564} \end{array}$

Therefore, the answer is 8 kg 94 g.

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Example 5

Divide 4 L 640 mL by 8

Steps	Solution
1. Arrange units of volume in column	L mL 8)4 640
2. Divide by considering the relationship between metric units of volume 3. Divide 4 L by 8. It is not sufficient, thus, write 0 in the L column 4. Convert 4 L into mL 5. Remember, $1\ \text{L} = 1000\ \text{mL}$, therefore 4L is equal to 4,000 mL 6. Add 4,000 mL to 640 mL to get 4,640 mL 7. Divide 4,640 mL by 8 to get 580 mL 8. Write 580 in the mL column	L mL 0 580 8)4 640

Therefore, the answer is 580 mL.

Exercise 4

1. m cm mm
   6) 6 56 60

2. m dm cm
   9) 3 65 22

3. km hm dam
   20) 5 200 20

4. km dam m
   8) 18 45 230

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5. 11 | km hm dam : 6 4 9

6. 6 | dm cm mm : 6 10 2

7. 2 | t kg g : 8 20 128

8. 9 | kg g : 7 200

9. 3 | t kg g : 6 4 20

10. 2 | t kg g : 15 245 26

11. 9 | t kg : 28 251

12. 17 | kg g : 34 850

13. 21 | t kg g : 2 100 63

14. 50 | kg g : 120 800

15. 14 | L mL : 30 520

16. 25 | L mL : 101 250

17. 6 | L mL : 720 198

18. The book shelf is 2 m and 10 cm long. A shelf consists of 15 books of the same thickness. Find the thickness of each book in m, dm and cm.

19. The total height of 235 children is 282 m. If all children have the same height, find the height of each child in m and dm.

20. The total width of 24 houses of the same width is 1,644 m. What is the width of each house in m and cm?

21. A total of 24,450 litres of milk were equally shared among 150 villages. How many litres of milk were received by each village?

22. A total of 85,995 kilograms of meat were equally shared among 105 schools. How many kilograms did each school get?

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23. Two hundred bags of groundnuts weighs 16,000 kg. Find the mass of each bag if all the bags have the same mass.
24. A total of 1,520 tonnes of finger millet were carried by 76 lorries. If each lorry carried the same mass, how many tonnes of finger millet were carried by each lorry?
25. Thirty five residents of a certain town are supposed to share 250 litres and 740 millilitres of water equally. How many litres of water will each resident get?

Activity 4 Measurements of length, mass and volume through online lessons

Descriptions

Use various programs such as Khan Academy, videos showing real examples that provide lessons and practical exercises about measurements.

Summary

1. In order to multiply or divide metric units, it is important to know the relationship among the units. Example, $1\ \text{km} = 1000\ \text{m}$, $1\ \text{kg} = 1000\ \text{g}$ and $1\ \text{L} = 1000\ \text{mL}$.
2. When multiplying or dividing metric units, the simplest way is to arrange the units in columns and then multiply or divide by the given number.
3. When multiplying metric units, start multiplying from the smallest unit on the right to the largest unit on the left.
4. When dividing metric units, start dividing from the largest unit on the left to the smallest unit on the right.

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Revision exercise

1.

m	cm
25	20
×	3

2.

m	cm	mm
25	20	50
×		3

3.

m	cm
21	38
×	4

4.

hm	m
13	50
×	3

5.

kg	dag	g
3	16	42
×		9

6.

dag	g
51	7
×	8

7.

kg	g
11	428
×	26

8.

kg	dag	g
1	62	5
×		4

9.

L	mL
6	12
×	8

10.

L	mL
8	14
×	17

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11. 4 | m: 8 | cm: 14 | mm: 24

12. 9 | dam: 48 | cm: 51

13. 2 | hm: 7 | dam: 8 | cm: 52

14 20 | km: 10 | m: 120

15. 3 | t: 4 | kg: 29 | g: 123

16. 23 | kg: 7 | g: 475

17. 5 | kg: 46 | dag: 5

18. 2 | t: 15 | kg: 245 | g: 26

19. 3 | L: 33 | mL: 540

20. 50 | L: 24 | mL: 800

21. 14 | L: 9 | mL: 184

22. 7 | L: 12 | mL: 110

23. A brick has a length of 12 centimetres and 30 millimetres. What is the total length of 50 bricks of the same kind?

24. A bunch of onions weighs 1 kilogram and 500 grams. What will be the mass of 30 bunches of onions of the same type in kg and g?

25. A water supply company sells an average of 5,500 litres and 760 millilitres of water per day. How many litres of water will this company sell in 30 days?

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26. A container with goods weighs 29 tonnes and 700 kilograms without its mass. If the container carries 45 items of the same kind, what is the mass of each item?

27. A farm has length 225 m and 90 cm and width 50 m. A farm is to be shared equally among 5 people considering the length. What length in metres of the portion will each person get?

28. A milk processing company supplied 2,500 litres and 500 millilitres of milk to 250 villagers. If the milk was supplied equally among the villagers, how much milk did each villager get?

29. A total of 60,000 kilograms of beans were equally shared among 25 secondary schools. How many kilograms of beans were received by each school?

30. 125 kilograms of rice were distributed equally to 8 people. How much rice in kilogram and gram, did each person get?

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Chapter Two

Circumference and area of a circle

Introduction

In this chapter, you will learn how to find the circumference and area of a circle. Additionally, you will learn how to deduce the concept of Pi. The competencies developed will enable you to deduce and apply the concepts of Pi, circumference and area of a circle in construction, engineering, as well as in tailoring and other daily life applications.

Think

Mathematics without knowledge of figures

Concept of Pi

Pi is a special number used in calculations involving circles. It is written in symbol as π and it represents the ratio between the circumference of a circle and its diameter.

Activity 1 **Finding the approximate value of Pi**

Steps

1. Take any circular object such as a plate, a cup, a bucket or a lid.
2. Wrap a measuring tape or a piece of rope around the edge of the object.

3. If using a rope, mark the point where it completes one full round, then measure its length using a ruler. This measurement represents the circumference of the object.
4. Mark the centre of the circular object and measure the diameter from one side of the circular object to the other side through the centre using a ruler.
5. Divide the circumference of the circular object in step 3 by its diameter obtained in step 4.
6. Repeat procedures 1 to 5 using different circular objects with different diameters and tabulate your results as follows:

Object	Circumference (cm)	Diameter (cm)	Circumference ÷ Diameter
Cup			
Plate			
Bucket			
Lid			

7. Does the ratio of the circumference to diameter in Step 6 remain the same for different circular objects?

If you measure properly, the results obtained after dividing the circumference by the diameter is approximately 3.141593. This ratio is known as Pi, denoted by the symbol π and is approximated as 3.14 in 2 decimal places. Also, as a fraction, Pi can be approximated as $\pi = \frac{22}{7}$.

Pi is a very important mathematical constant used primarily in calculating circumferences and areas of a circles. Knowing its value increase more understanding about circular objects in daily life.

$$\text{Therefore, Pi} = \frac{\text{Circumference of circular object}}{\text{Diameter of the circular object}}$$

Circumference of a circle

The circumference of a circle is the distance around the circle. This is the length of a string that goes around a circle from the start to the end.

For example, consider the bicycle tyre ABCDA whose centre is at point O. Circumference of the bicycle tyre is the distance sorounding bicycle tyre. This is the length of rope sorounding the bicycle tyre from point A, B, C, D turning to point A.

$\overline{AC}$ is the diameter of the circle of the bicycle tyre and the distance from the centre of the circle of the bicycle tyre to any point on the circumference of a circle of the bicycle tyre is called the radius. Thus, $\overline{OB}$, $\overline{OA}$ and $\overline{OC}$ are the radii of the circle of the bicycle tyre.

The diameter of a circle is denoted by a letter d, and its radius is denoted by a letter r. Hence, $d = 2r$.

Given the diameter or radius of the circular object, the circumference of a circular object is therefore given by $C = \pi d$ or $C = 2\pi r$ where C is a circumference of the circular object.

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Activity 2 — Using Pi to measure the circumference of a circle

Steps

1. Use a measuring tape to measure and record the diameter of a bicycle tyre.
2. Calculate the circumference of the bicycle tyre using the formula: Circumference $= \pi d$. Where, $\pi = \frac{22}{7}$.
3. Use a chalk to mark a starting point of the bicycle tyre.
4. Wrap the rope around the tyre from one sign until when you reach that mark again to get one complete circle.
5. Measure the length of the rope you have used in step 4.
6. Compare the calculated circumference in step 5 with the one you got in step 2. What is your conclusion?

Thus, circumference of a circle $= \pi \times d$.

Remark: The circumference of the semi-circle can be found by taking half of circumference of the circle plus the diameter.

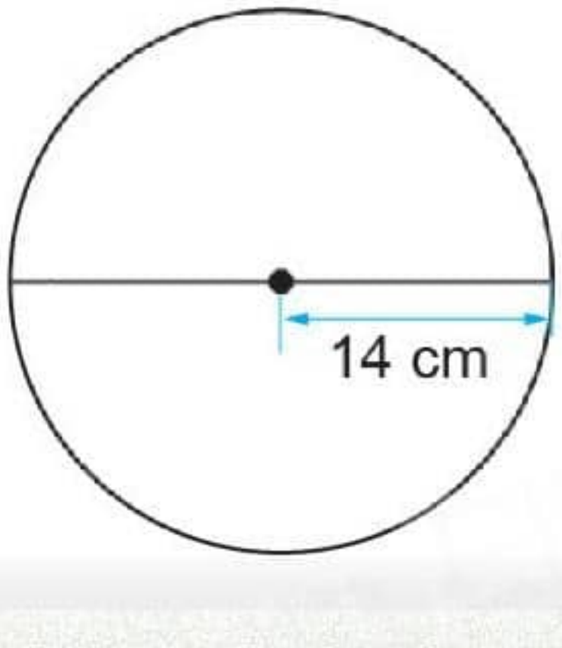
That is, the circumference of the semi-circle $= \frac{\pi d}{2} + d$.

Example 1

Find the circumference of a circle with a radius of 14 cm.

Use $\pi = \frac{22}{7}$.

Solution



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$$C = 2\pi r$$

$$= 2 \times \frac{22}{7} \times 14 \text{ cm}$$

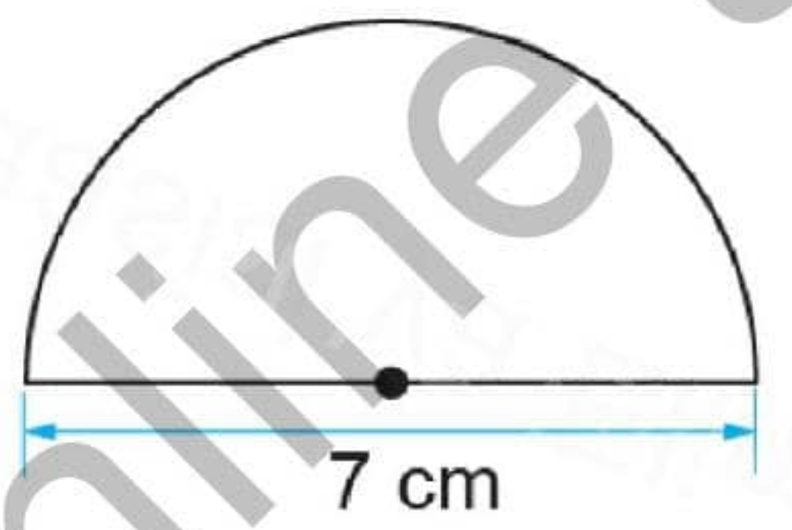
$$= 88 \text{ cm}$$

Therefore, the circumference of the circle is 88 cm.

Example 2

Find the circumference of a semi-circular figure with a diameter of 7 cm. Use $\pi = \frac{22}{7}$.

Solution



$$\text{Circumference of a semi-circular figure} = \frac{\pi d}{2} + d$$

$$= \frac{22}{7} \times \frac{7}{2} \text{ cm} + 7 \text{ cm}$$

$$= 18 \text{ cm}$$

Therefore, the circumference of the semi-circular figure is 18 cm.

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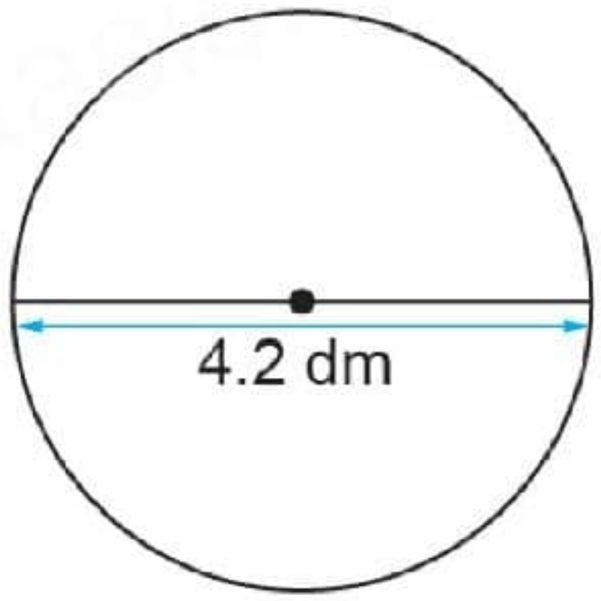
Example 3

Find the circumference of a circle whose diameter is 4.2 dm.

Use $\pi = \frac{22}{7}$.

Solution



$$C = \pi d$$

$$= \frac{22}{7} \times 4.2 \text{ dm}$$

$$= 13.2 \text{ dm}$$

Therefore, the circumference of the circle is 13.2 dm.

Example 4

Find the diameter of a circle whose circumference is 110 cm.

Solution

$C = \pi d$

Divide both sides by π to make d the subject,

$$\frac{C}{\pi} = \frac{\pi d}{\pi}$$

Thus, $d = \frac{C}{\pi}$

$$= 110 \text{ cm} \times \frac{7}{22}$$

$$= 35 \text{ cm}$$

Therefore, the diameter of the circle is 35 cm.

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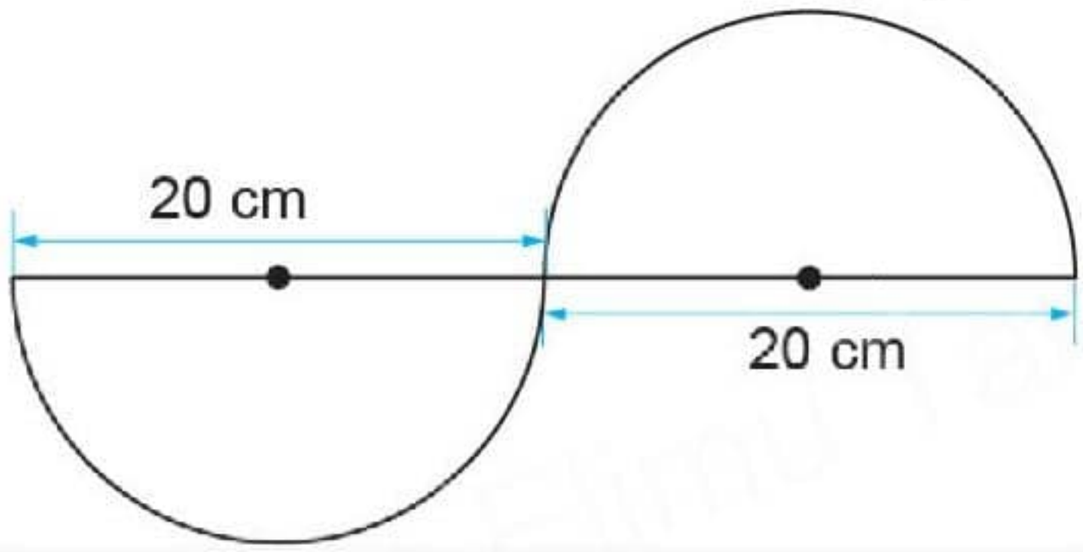
Exercise 1

Answer the following questions. Use $\pi = \frac{22}{7}$ unless stated otherwise:

1. Find the circumference of the circle using each of the following diameters:
   (a) 98 cm (b) 56 cm (c) 9.1 dm (d) 7 dm
2. Find the diameter of the circle for each of the given circumferences:
   (a) 44 m (b) 132 m (c) 220 mm
3. In each of the following circumferences, find the radius of each circle:
   (a) 968 dm (b) 616 dm
4. Find the circumference of three-quarters circle whose radius is 35 cm.
5. Find the circumference of each circle with the following measurements:
   (a) $\text{diameter} = 80.5$ cm (b) $\text{radius} = 210$ mm
6. The diameter of a car tyre is 63 cm. If the tyre rotates to cover a distance of 63.36 metres, how many complete rotations will the tyre make?
7. Mr Baraka constructed a well with a diameter 7 m. During the early stages of its construction, there were pillars placed 2.2 m apart to strengthen the well. How many pillars were used in the construction of the well?
8. Find the circumference of the following figure. Use $\pi = 3.14$.



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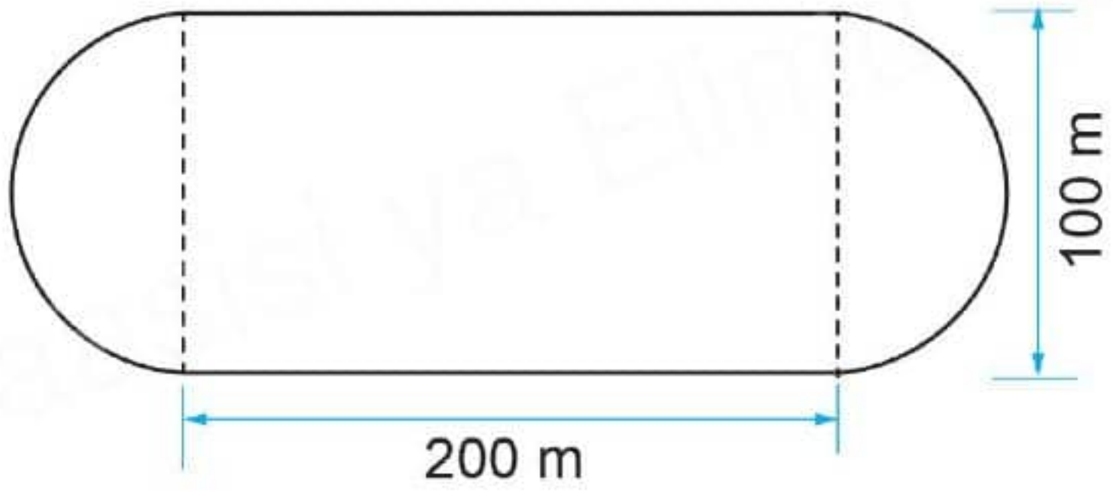
9. Find the circumferences of the following playing ground. Use $\pi = 3.14$.



10. Find the circumference of the semi-circular figures with the following diameters:
    (a) 21 m (b) 84 cm
11. Find the circumference of a quarter circle with a radius 49 cm.
12. The circumference of the circular base of a bowl is 110 mm. Find its diameter.

Area of a circle

In this section, you will learn how to find an area of a circle.

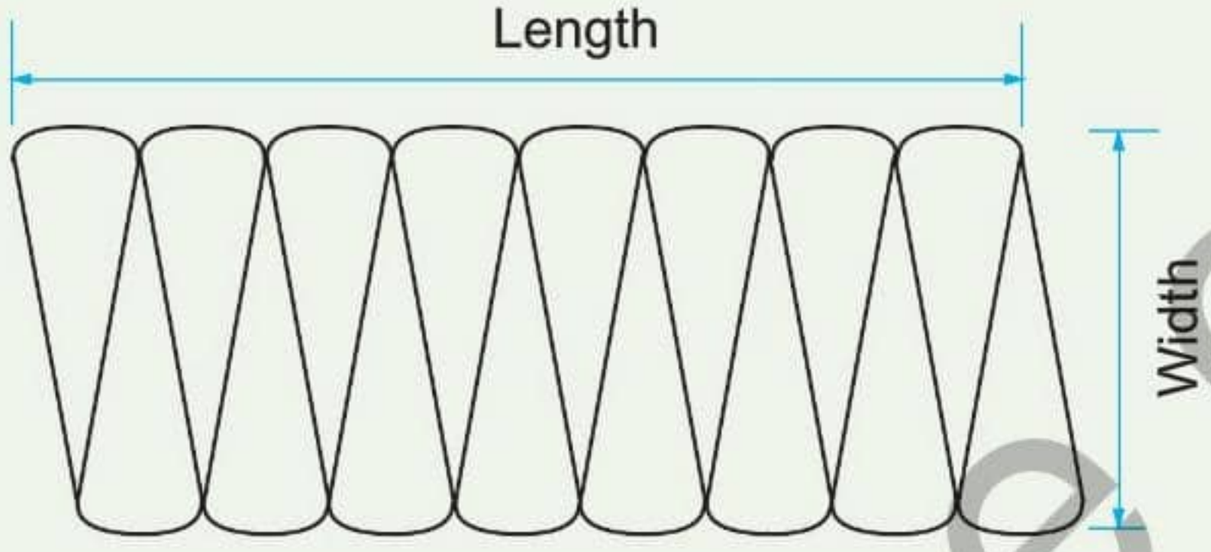
Activity 3 Finding the area of a circle

Make sure you have a manila paper or other piece of paper, a razor blade or a pair of scissors, a pen or pencil and a pair of compasses.

Steps

1. Use the pair of compasses to draw a circle on a manila paper.
2. Cut the circular plane shape out of the manila card. Divide it into 16 equal parts as shown in the following figure:

3. Measure the diameter and use it to calculate the circumference of the circle.
4. Cut carefully all the parts by using a pair of scissors or a razor blade and then arrange the parts to form a rectangular shape as shown in the following figure:



5. Measure the length and width of the rectangular shape and record answers in your exercise book.
6. Divide the calculated circumference of the circle in Step 3 by 2 and then compare your answer with the length of the rectangle. What have you discovered?
7. Divide the circumference of the original circle by 2 and compare the obtained value with the length of the rectangle. What have you discovered?
8. Find the area of the rectangular shape by using the length and width you have obtained.

You might have discovered that the length of the rectangle you constructed is half the circumference of the circle.

Therefore, length of the rectangular shape $= \frac{C}{2} = \frac{2\pi r}{2} = \pi r$

Also, the width of the rectangle is equal to the radius of the circle.

Therefore, the area of a circle can be obtained by calculating the area of the rectangle constructed. This is possible because the length and width of the rectangle constructed are known. Therefore, you can write the formulae for the area of a circle as follows:

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The area of a circle = Area of rectangular shape

$$= \text{length} \times \text{width}$$

$$= \pi r \times r$$

$$= \pi r^2$$

Therefore, the area of a circle is πr^2 where r is the radius of the circle.

Now, use the formula with $\pi = 3.14$ to calculate the area of the circle, then, compare the answer with the area of the rectangular shape you have constructed. The answers are expected to be approximately the same.

Since, diameter (d) $=$ $2 \times$ radius (r)

$$d = 2r$$

$$r = \frac{d}{2}$$

Thus, area of a circle $= \pi r^2 = \frac{\pi d^2}{4}$

Therefore, the area of a circle is given by $\frac{\pi d^2}{4}$ or πr^2 where r is the radius and d is the diameter.

Example 1

Find the area of a circle with radius 14 mm. Use $\pi = \frac{22}{7}$

Solution

Area of a circle $= \pi r^2$

$$= \frac{22}{7} \times 14\,\text{mm} \times 14\,\text{mm}$$

$$= 616\,\text{mm}^2$$

Therefore, the area of the circle is $616\,\text{mm}^2$.

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Example 2

Find the area of a circle with diameter 21dm.

Solution

$$\text{Area of a circle} = \pi \frac{d^2}{4}$$

$$= \frac{22}{7} \times \frac{21\,\text{dm} \times 21\,\text{dm}}{4}$$

$$= 346.5\,\text{dm}^2$$

Therefore, the area of the circle is $346.5\ \text{dm}^2$

Example 3

Given that the circumference of the circle is 88 m, find the area of the circle.

Solution

$$\text{Circumference} = \pi d$$

$$88 = \frac{22d}{7}$$

$$d = 28\ \text{m}$$

$$\text{Area of the circle} = \frac{\pi d^2}{4}$$

$$= \frac{22}{7} \times \frac{28 \times 28}{4}$$

$$= 616\ \text{m}^2$$

Therefore, the area of the circle is $616\ \text{m}^2$.

Alternatively

$$\text{Area of the circle} = (\text{circumference} \times \text{circumference}) \div \pi \times 4$$

$$= 88 \times 88 \div \frac{22}{7} \times 4$$

$$= 88 \times 88 \times \frac{22}{88}$$

$$= 616\ \text{m}$$

Therefore, the area of the circle is $616\ \text{m}^2$.

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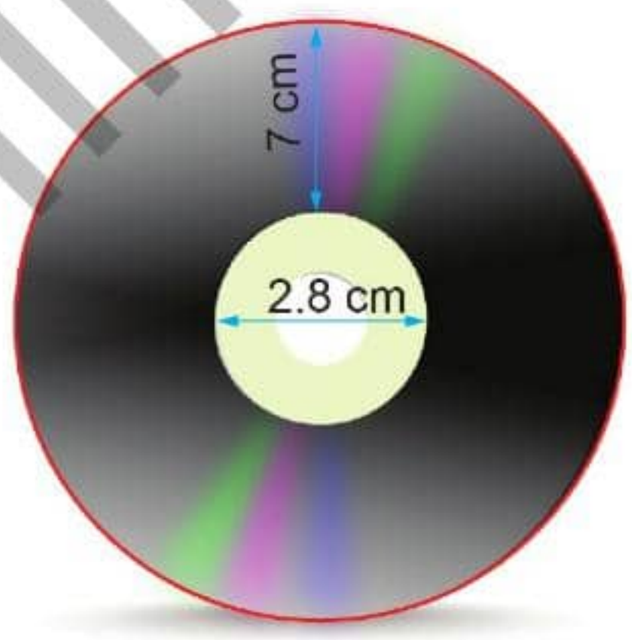
Exercise 2

1. For each of the following radii, find the area of the circle.
   (a) 80 mm (b) 2 dm (c) 3.5 cm (d) 2.52 m
2. Use each of the following diameters to find the area of the circle. Use $\pi = 3.14$.
   (a) 8 m (b) 6 cm (c) 2 dm (d) 3.5 cm (e) 22 dm (f) 3 cm
3. Find the area of the opening part of a circular-shaped fish pond having a diameter of 98 m.
4. Two circular playgrounds have radii of 6 m and 8 m, respectively. Without calculating exact values of their areas, which playground has larger area? Why?
5. Find the area of a circle with a radius of 8 cm. Use $\pi = 3.14$.
6. A bucket lid has a radius of 16 cm. Find its area. Use $\pi = 3.14$.
7. Compare the area of a square with a side length of 10 cm to the area of a circle with the radius of 10 cm. Which area is larger? By how much?
8. In the following disc, find the area of the shaded region. Use $\pi = 3.14$.



9. If the circumference of a circle is 88 dm, find its area.
10. A villager built a circular house of a diameter 28m. The mason constructed the floor at a cost of 5,000 shillings per square metre. How much money did the villager pay the mason? Use $\pi = \frac{22}{7}$.

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Activity 4 Learning the area and circumference of a circle through online lessons

Descriptions

Use various online programs such as Khan Academy to get video lessons and practical exercises about the area and circumference of a circle.

Summary

1. The value of π is 3.141593…, which is approximated to 3.14 to 2 decimal places, and in fraction, π can also be approximated as $\frac{22}{7}$.
2. The circumference of a circle with a diameter d is given by: Circumference $= \pi d$.
3. The circumference of a circle with a radius r is given by: Circumference $= 2\pi r$.
4. The circumference of a semi-circle with a diameter d is given by: $\frac{\pi d}{2} + d$.
5. Area of a circle $= \frac{\pi d^2}{4}$ or πr^2.

Revision exercise

1. Consider the following figure:
   (a) find its circumference.
   (b) find its area.

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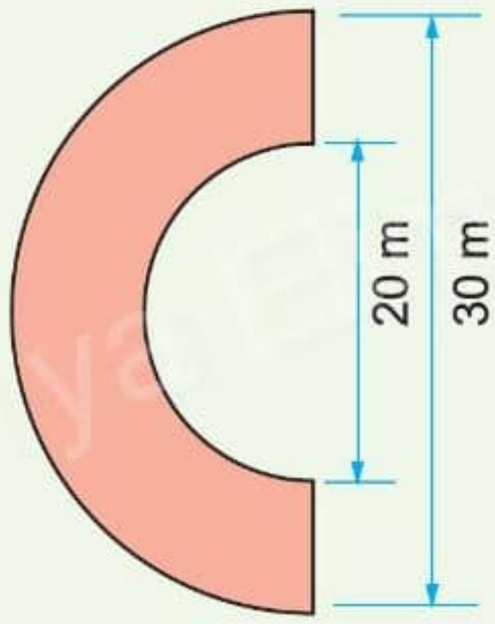
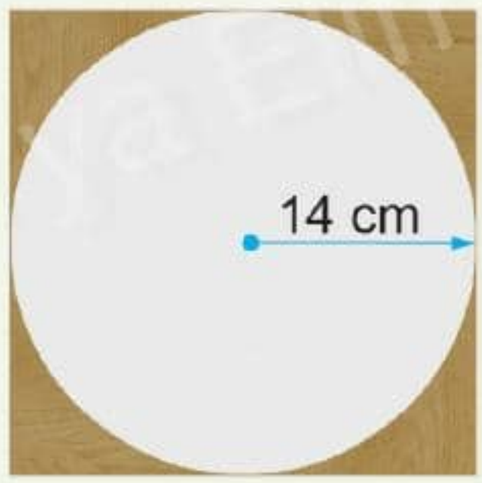


2. The circumference of a quarter-circular is 50 cm. Find its area.
3. If the diameter of a circle is 7 cm, find its area. Use $\pi = 3.14$
4. A student measures a plate and finds the diameter to be 20 cm. Find its circumference.
5. If the circumference of a wheel is 62.8 cm, find its diameter. Use $\pi = 3.14$.
6. James has a round plate with a diameter of 12 cm. Find the radius of the plate.
7. A circular clock has a radius of 4 cm. Find its area and circumference. Use $\pi = 3.14$.
8. Find the area of the shaded part in the following figure. Use $\pi = 3.14$.



9. The circumference of a quarter circle is 11 cm. Find its radius. Use $\pi = 3.14$.
10. The circumference of a three quarter circle is 37.68 m, find its radius.

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Chapter Three

Types, factors and multiples of numbers

Introduction

In this chapter, you will learn about different types of numbers, namely, even, odd and prime numbers. You will also learn about factors and multiples of numbers. You will learn about factors and multiples in finding the Greatest Common Factor *(GCF) and the Lowest Common Multiple (LCM) of whole numbers. The competencies developed will enable you in comparing prices, exchanging money, dividing things into equal parts, understanding time and measuring quantities.*

Think

Life without numbers

Types of numbers

Numbers can be categorized into several types depending on its use. A number can be natural or counting, whole, even, odd, prime and many others. In this section you will learn about the concept of even, odd and prime numbers.

Even numbers

Even numbers are whole numbers which are exactly divisible by 2. The last digit of an even number is either 0, 2, 4, 6 or 8. Zero is also an even number.

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Activity 1 — Identifying even numbers from 1 up to 12 by using counters

Steps

1. List the counting numbers from 1 up to 12.
2. Use counters to represent the numbers.
3. Put the counters into groups of two, then circle each group.
4. In step 3 you observed that, there are numbers with groups of exactly two counters with no remainder.
5. Write the numbers identified in step 4.

The numbers obtained are called even numbers.

Therefore, even numbers are whole numbers which are divisible by 2 without a remainder.

Example 1

List all even numbers between 17 and 33.

Answer

The even numbers between 17 and 33 are 18, 20, 22, 24, 26, 28, 30 and 32.

Example 2

List only missing numbers that are even in the following pattern:

574, _____, _____, _____, _____, _____, _____, 581.

Answer

The missing even numbers in the given patten are

576, 578 and 580.

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Exercise 1

1. List all even numbers between:

   (a) 25 and 35 (b) 50 and 60 (c) 81 and 99

2. Identify and write all even numbers from the following list:

   (a) 43, 52, 56, 59, 60

   (b) 71, 78, 82, 83, 87

   (c) 92, 97, 96, 98, 99

3. Which among the following numbers are not even?

   (a) 52, 55, 56, 57, 59, 60, 62

   (b) 18, 33, 36, 37, 40, 58, 70

4. Fill in the blanks in the following pattern:

   22, 24, 26, ____, ____, ____.

5. How many even numbers are there between 45 and 63?

6. Write all even numbers between 73 and 93 which are exactly divisible by 4.

7. Write the missing even numbers in the following pattern:

   5, 10, 15, ____, ____, ____, ____, ____, ____, ____.

8. Write even numbers from 4 up to 10.

Odd numbers

Odd numbers are the counting numbers which are divisible by two with a remainder 1.

Activity 2 — Identifying odd numbers from 1 up to 20 by using counters

Steps

1. List the counting numbers from 1 up to 20.
2. Represent the numbers by counters.
3. Put the counters into groups of two, then circle each group.
4. In step 3, you observed that there is a number with exactly 1 counter and others with groups of two counters and a remainder of 1 counter.
5. Write the numbers obtained in step 4.

The numbers obtained are called odd numbers.

Example 1

Write all odd numbers from the following list: 38, 49, 87, 96, 105, 254 and 281.

Answer

The odd numbers from the given list are: 49, 87, 105 and 281.

Example 2

Write only missing odd numbers in the following pattern:

1003, 1006, 1009, _____, _____, _____, _____.

Answer

The missing odd numbers are: 1015 and 1021.

Exercise 2

1. Given the following list of numbers:
   4, 5, 8, 11, 13, 14, 16, 19, 21, 22, 25, 27, 29, 31, 32, 34, 35.
   (a) Write all odd numbers.
   (b) Write all odd numbers between 8 and 22.
2. Write down only missing odd numbers in the following pattern:
   9, 12, 15, ___, 21, ___, ___, ___, 33, 36, ___, ___.
3. How many odd numbers are there in the first 10 counting numbers?
4. Write odd numbers from 13 to 31.
5. List all odd numbers between 31 and 50 that are exactly divisible by 5.

Prime numbers

A prime number is a counting number which is divisible by 1 and itself. One is not a prime number since it has only one factor, that is itself.

Activity 3 — Identifying all prime numbers between 1 up to 100

Steps

1. List the counting numbers from 1 up to 100.
2. Encircle number 2, then mark all numbers divisible by 2 with X.
3. Encircle number 3, then mark all numbers divisible by 3 with X.
4. Encircle number 5, then mark all numbers divisible by 5 and have not been marked with X.

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5. Encircle number 7, then mark all numbers divisible by 7 and have not been marked with X.
6. Write the numbers which have been encircled and those which have not been marked with X except 1.
7. What did you discover in step 6.

Each obtained number is divisible by 1 and itself and is called a prime number.

Example 1

List all prime numbers less than 20.

Answer

The prime numbers which are less than 20 are:
2, 3, 5, 7, 11, 13, 17 and 19.

Example 2

Write all prime numbers between 101 and 120.

Steps

1. List all numbers between 101 and 120 which are:

   102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118 and 119.
2. Mark all numbers that are divisible by 2 with X. The remaining numbers are 103, 105, 107, 109, 111, 113, 115, 117 and 119.
3. Mark all numbers that are divisible by 3 with X. The remaining numbers are 103, 107, 109, 113, 115 and 119.
4. Mark all numbers that are divisible by 5 with X. There is no any number divisible by 5.

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5. Mark all numbers that are divisible by 7 with X. The remaining unmarked numbers are 103, 107, 109 and 113.

Thus, the prime numbers between 101 and 120 are: 103, 107, 109 and 113.

Example 3

Write the prime numbers in the following list:

19, 21, 29, 37, 44, 45, 49, 51, 55, 57, 59.

Answer

The prime numbers are:

19, 29, 37 and 59.

Exercise 3

1. Write all the prime numbers in the following list:
   4, 7, 10, 12, 13, 15, 16, 17.
2. List three prime numbers between 14 and 25.
3. Write a prime number which is also an even number.
4. How many prime numbers are there between 10 and 20?
5. I have a prime number from which when I substract an even number the answer is zero. What number is that?
6. List all prime numbers between 30 and 80.
7. Write the missing prime numbers in the following pattern:
   31, _____, ____, _____, 47.
8. Write all the prime numbers from the following list:
   80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94 and 95.
9. Write the sum of prime numbers from 19 to 31.

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Factor of a number

A factor of a given number is any whole number which divides that number without a remainder. The plural of a factor is factors.

Example 1

Find all factors of 6.

Solution

Divide 6 by 1, 2, 3, 4, 5 and 6.

$6 \div 1 = 6$, $6 \div 4 = 1$ remainder 2,

$6 \div 2 = 3$, $6 \div 5 = 1$ remainder 1,

$6 \div 3 = 2$, $6 \div 6 = 1$.

The numbers which divide 6 without remainder are 1, 2, 3 and 6.

Therefore, the factors of 6 are 1, 2, 3 and 6.

Example 2

List all factors of 36 which are divisible by 2.

Solution

Find all the factors of 36.

$36 \div 1 = 36$, $36 \div 9 = 4$,

$36 \div 2 = 18$, $36 \div 12 = 3$,

$36 \div 3 = 12$, $36 \div 18 = 2$,

$36 \div 4 = 9$, $36 \div 36 = 1$.

$36 \div 6 = 6$,

The factors of 36 are: 1, 2, 3, 4, 6, 9, 12, 18 and 36.

The factors of 36 which are divisible by 2 are: 2, 4, 6, 12, 18 and 36.

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Example 3

Find all the factors of 45.

Solution

Express 45 as a product of two numbers using different numbers:

$1 \times 45 = 45$, 1 and 45 are factors of 45.

$3 \times 15 = 45$, 3 and 15 are factors of 45.

$5 \times 9 = 45$, 5 and 9 are factors of 45.

Therefore, the factors of 45 are 1, 3, 5, 9, 15, and 45.

Exercise 4

1. Write all the factors of the following numbers:

   (a) 20 (b) 7 (c) 12 (d) 5

2. Find all the factors of the following numbers:

   (a) 8 (b) 16 (c) 24 (d) 49 (e) 10 (f) 81

3. Fill in the boxes with the missing factor:

   (a) $40 = 2 \times \square \times 5$ (b) $72 = 1 \times 2 \times 3 \times \square$

   (c) $100 = 2 \times \square \times 10$ (d) $90 = 3 \times \square \times 6$

   (e) $56 = \square \times 4 \times 7$

4. (a) $120 \div \square = 20$ (b) $\square \div 5 = 30$ (c) $160 \div \square = 80$

Finding factors of a number by using prime numbers

Prime numbers can also be identified by using tree diagrams.

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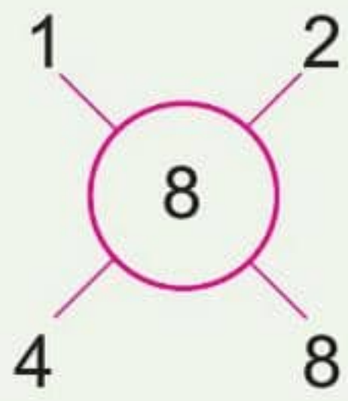
Activity 4 Identifying prime numbers by using factors

Steps

1. List all counting numbers from 1 to 15 and encircle them:
2. Write factors of each number and represent them using branches. For example;



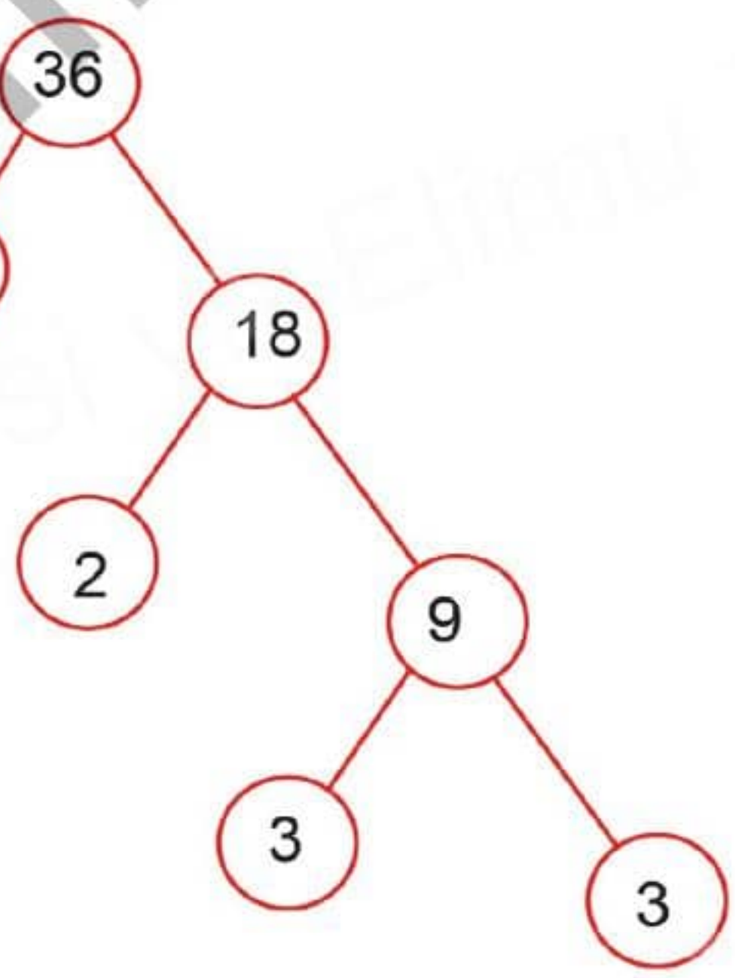
3. List all numbers which have only two branches, that is, numbers with only two factors.

The numbers obtained in step 3 are called prime numbers.

Example 1

Write 36 as a product of prime factors.

Solution



Therefore, $36 = 2 \times 2 \times 3 \times 3$.

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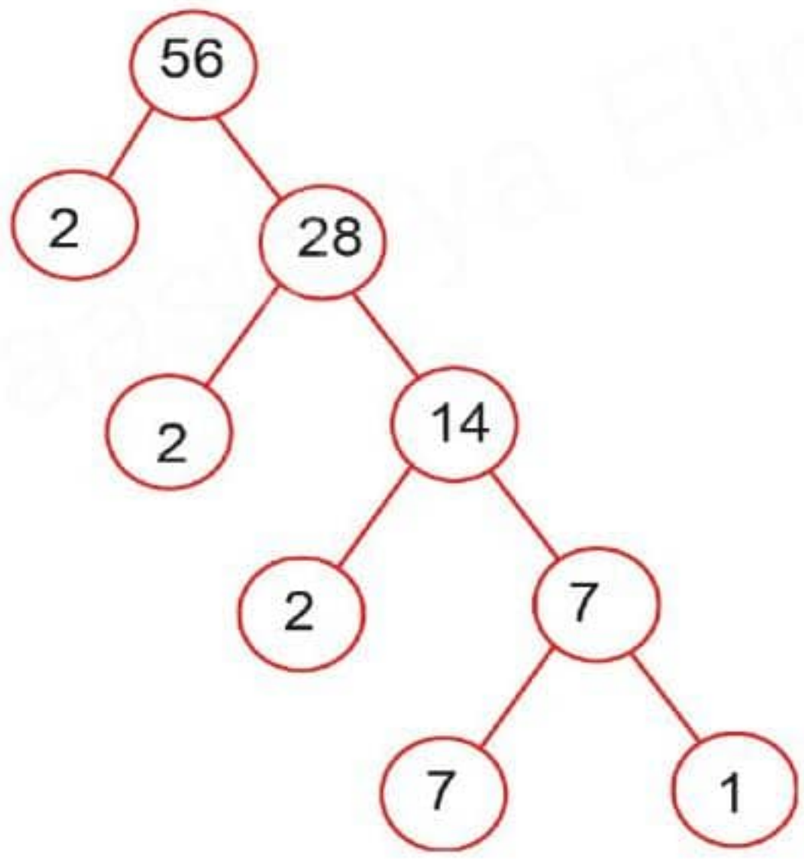
Example 2

Find all prime factors of 56.

Solution



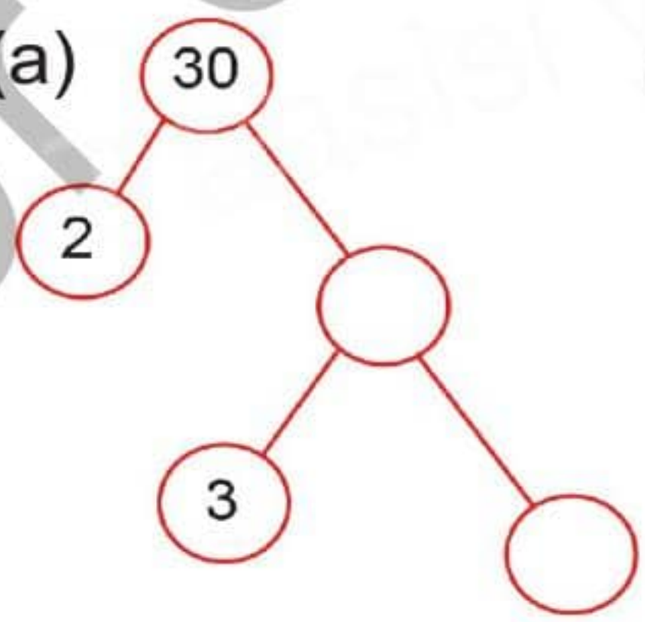
The prime factors of 56 are 2 and 7

Therefore, $56 = 2 \times 2 \times 2 \times 7$

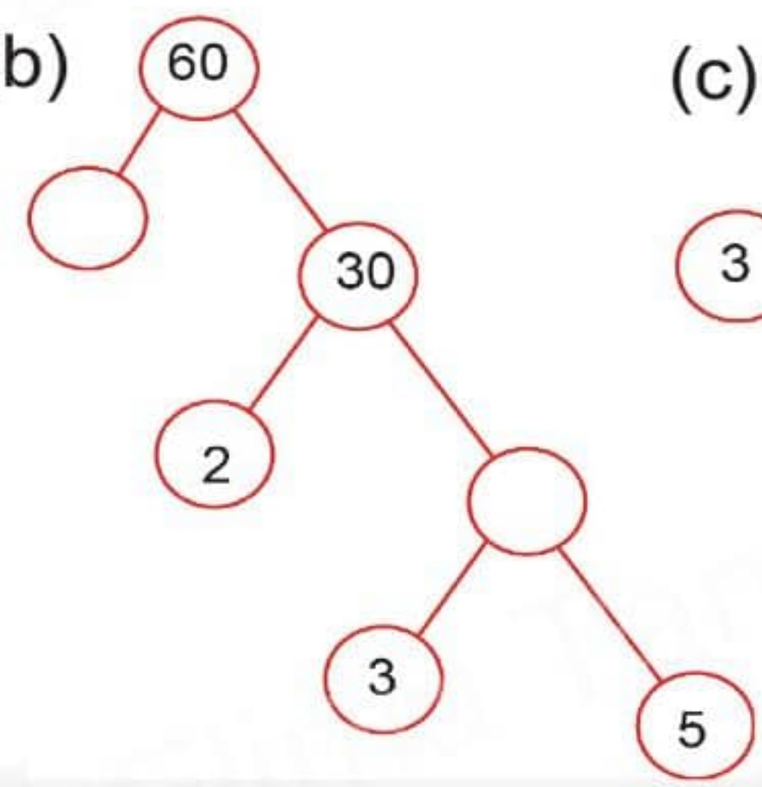
Exercise 5

1. Find the prime factors of the following numbers:
   (a) 12 (b) 28 (c) 15 (d) 24 (e) 14 (f) 154
2. How many prime factors does 9 have?
3. List all prime factors of 100.
4. Write 45 as a product of prime factors.
5. Fill the missing factors in the following tree diagrams:

(a)



(b)



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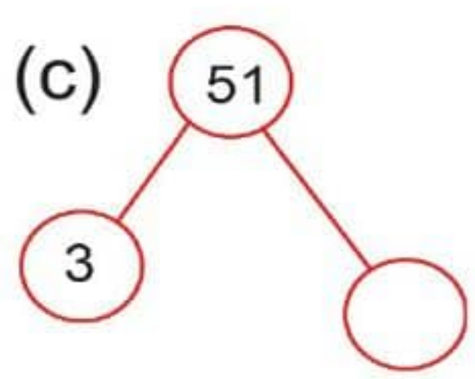

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Greatest Common Factor (GCF)

The Greatest Common Factor (GCF) is the largest number that divides two or more numbers without a remainder.

Example 1

Find the common factors of 6 and 8.

Solution

List the factors of 6 and 8:

The factors of 6 are 1, 2, 3 and 6.

The factors of 8 are 1, 2, 4 and 8.

Therefore, the common factors of 6 and 8 are 1 and 2.

Example 2

Use the listing method to find the GCF of 30 and 42.

Solution

List the factors of 30 and 42.

The factors of 30 are 1, 2, 3, 5, 6, 10, 15, and 30.

The factors of 42 are 1, 2, 3, 6, 7, 14, 21, and 42.

The common factors of 30 and 42 are 1, 2, 3 and 6.

The greatest common factor of 30 and 42 is 6.

Therefore, the GCF of 30 and 42 is 6.

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Example 3

Find the GCF of 12 and 18 by using prime factors.

Solution

Express 12 as a product of prime factors: $12 = 2 \times 2 \times 3$.

Express 18 as a product of prime factors: $18 = 2 \times 3 \times 3$.

The common prime factors of both numbers are 2 and 3.
The product of 2 and 3 is 6.

The product of common prime factors is the GCF.

Therefore, the GCF of 12 and 18 is 6.

Example 4

Find the GCF of 24 and 60 by using common prime factors.

Solution

2	24	60	24 and 60 are divisible by 2
2	12	30	12 and 30 are divisible by 2
2	6	15	2 is a factor of 6 but not of 15
3	3	15	3 and 15 are divisible by 3
5	1	5	5 is divisible by 5
	1	1	when 1 is reached, the process ends

The prime factors that divide both numbers without a remainder in that sequence are 2 and 3. The common prime factors of 24 and 60 are 2, 2 and 3.

Thus, $2 \times 2 \times 3 = 12$

Therefore, the GCF of 24 and 60 is 12.

Exercise 6

1. Use listing method to find the factors of the following numbers and identify the GCF.
   (a) 10 and 12
   (b) 8 and 16
   (c) 15 and 27
   (d) 96 and 36
   (e) 42 and 72
   (f) 7 and 15
   (g) 20 and 22
   (h) 18 and 45

2. Use prime factors to find the GCF of the following pairs of numbers:
   (a) 12 and 42
   (b) 15 and 20
   (c) 30 and 90
   (d) 8 and 24
   (e) 4 and 8
   (f) 16 and 24
   (g) 12 and 16

3. Use the method of tree diagram to find the GCF of the following pairs of numbers:
   (a) 42 and 48
   (b) 33 and 72
   (c) 72 and 144
   (d) 75 and 80

4. Use common prime factors to find the GCF of the following pairs of numbers.
   (a) 23 and 5
   (b) 24 and 156
   (c) 24 and 180
   (d) 35 and 245
   (e) 36 and 162

Multiples of numbers

If any number is multiplied by counting number, then the result is called a multiple. The plural of multiple is multiples.

$12 \times 1 = 12,$

$12 \times 2 = 24,$

$12 \times 3 = 36,$

Therefore 12, 24 and 36 are multiples of 12.

Multiples of a number can be listed by using the following examples.

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Example 1

Write the first five multiples of 2.

Solution

In order to get the first five multiples of 2, multiply 2 by 1, then by 2, 3, 4 and 5 as follows:

$2 \times 1 = 2,$

$2 \times 2 = 4,$

$2 \times 3 = 6,$

$2 \times 4 = 8,$

$2 \times 5 = 10.$

Therefore, the first five multiples of 2 are 2, 4, 6, 8 and 10.

Example 2

Write the multiples of 5.

Solution

The multiples of 5 are obtained by multiplying 5 by 1, then by 2, 3, 4 and so on as follows:

$5 \times 1 = 5,$ $5 \times 4 = 20,$

$5 \times 2 = 10,$ $5 \times 5 = 25,$

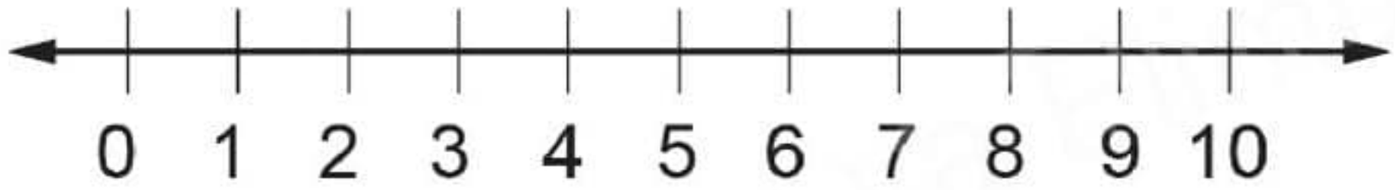
$5 \times 3 = 15,$ $5 \times 6 = 30.$

Therefore, the multiples of 5 are 5, 10, 15, 20, 25…

The list of multiples has no end unless a limiting value is given.

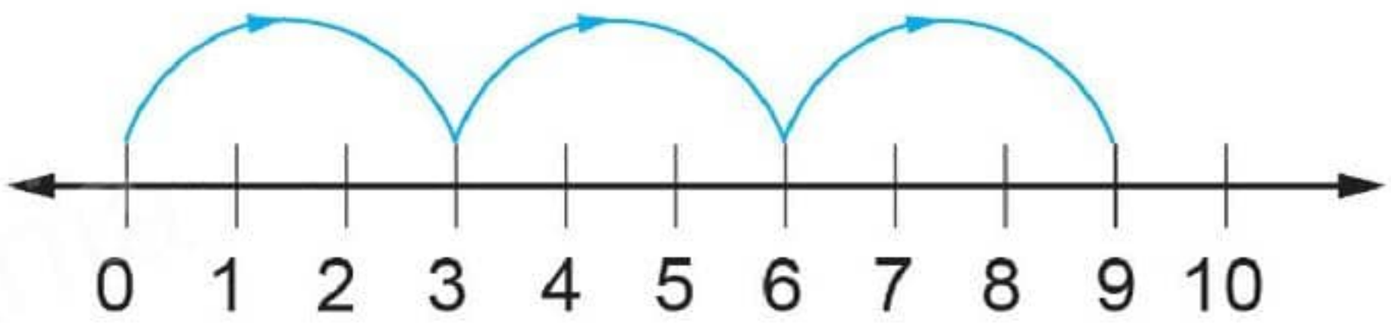
Example 3

Write down the multiples of 3 using the following figure.



Solution

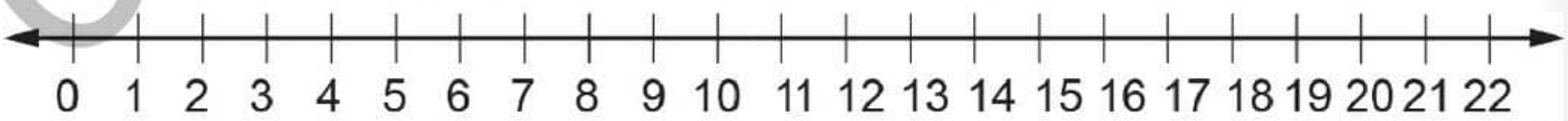
Starting from 0 move three successive steps.



Therefore, the multiples of 3 are 3, 6, 9, . . .

Exercise 7

1. Write all the multiples of 4 between 2 and 30.
2. Write all the multiples of 11 between 9 and 45.
3. List the first ten multiples of 9.
4. Write all the multiples of 7 between 45 and 100.
5. List all the multiples of the following numbers which do not exceed 60:
   (a) 5 (b) 8 (c) 10 (d) 17
6. List all multiples of 6 between 5 and 62.
7. Use the following figures to answer the questions that follow:



(a) circle all multiples of 5.

(b) mark all multiples of 2 by X.

(c) list the first 5 multiples of 1.

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Lowest Common Multiple (LCM)

The Lowest Common Multiple (LCM) of two or more numbers is the smallest whole number which is a multiple of the given numbers. It is the smallest common multiple of the given numbers.

Example 1

List the common multiples of 3 and 6.

Steps

1. Write the multiples of 3.
   The multiples of 3 are 3, 6, 9, 12, 15, 18, 21, 24, . . .
2. Write the multiples of 6.
   The multiples of 6 are 6,12,18, 24, 30, 36, 42, 48, …
3. Analyse the multiples of 3 that are also multiples of 6.
   The common multiples of 3 and 6 are 6,12,18, 24, …

   Therefore, the common multiples of 3 and 6 are: 6,12,18, 24, …

Example 2

Find the LCM of 3 and 4.

Steps

1. Write the multiples of 3.
   The multiples of 3 are 3, 6, 9, 12, 15, 18, 21, 24, 27, …
2. Write the multiples of 4.
   The multiples of 4 are 4, 8, 12, 16, 20, 24, 28, …
3. The common multiples of 3 and 4 are 12, 24, …
   Therefore, the LCM of 3 and 4 is 12.

Example 3

Use the method of prime factor divisors to find the LCM of 12 and 20.

Solution

2	12	20	12 and 20 are divisible by 2
2	6	10	6 and 10 are divisible by 2
3	3	5	3 divides only 3 but not 5
5	1	5	5 divides 5
	1	1	when 1 is reached, the process ends

$2\times2\times3\times5=60$

The LCM of 12 and 20 is the product of the prime factors 2, 2, 3, 5.

Therefore, the LCM of 12 and 20 is 60.

Example 4

Find the LCM of 8 and 30 using the method of prime factor divisors·

Solution

2	8	30
2	4	15
2	2	15
3	1	15
5	1	5
	1	1

$2\times2\times2\times3\times5=120$.

The Lowest Common Multiple of 8 and 30 is the product of the prime factors 2, 2, 3 and 5.

Therefore, the LCM of 8 and 30 is 120.

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Exercise 8

1. List all multiples of 9 between 20 and 60.
2. Write all multiples of 5 between 30 and 47.
3. List all common multiples of 5, 8 and 20 between 10 and 100.
4. List the first 5 multiples of each of the following numbers:

   (a) 6 (b) 35 (c) 13 (d) 24
5. Find the LCM of the following pairs of numbers:

   (a) 5 and 7 (b) 12 and 40 (c) 8 and 12
6. Find the LCM of the following pairs of numbers using product of prime factors.

   (a) 18 and 30 (b) 45 and 60 (c) 23 and 46

   (d) 25 and 85 (e) 150 and 225 (f) 96 and 128

Activity 5 Identifying types of numbers, factors and multiples of numbers through online lessons

Descriptions

Use various online programs such as Khan Academy to get video lessons and practical exercises on types, factors and multiples of numbers to deepen understanding.

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Summary

1. Even numbers are whole numbers that are exactly divisible by 2.
2. Odd numbers are counting numbers which are divisible by 2 with a remainder of 1.
3. A prime number is a counting number that has a exactly 2 distinct factors, which are 1 and itself.
4. 2 is the only number which is both even and prime.
5. Factors are numbers which divide a given number without a remainder.
6. Multiples are numbers which are divisible by the given number without a remainder.

Revision exercise

1. List all the factors for each of the following numbers:
   (a) 70 (b) 95 (c) 224 (d) 250 (e) 729
2. Write the prime factors for each of the following numbers:
   (a) 108 (b) 120 (c) 840
3. Write the common factors of each of the following pairs of numbers by listing.
   (a) 9 and 15 (b) 12 and 72 (c) 8 and 32
4. Find the common factors of each of the following pairs of numbers.
   (a) 60 and 200 (b) 28 and 32

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5. List the first 8 multiples for each of the following numbers:
   (a) 7 (b) 9 (c) 20 (d) 13 (e) 34

6. Find the LCM of 24 and 27 by using prime factors.

7. List all prime numbers between 30 and 40.

8. Write the type of a number which results from adding:
   (a) two even numbers
   (b) an odd number and even number
   (c) two odd numbers

9. Find the GCF of each of the following pairs of numbers:
   (a) (8, 12) (b) (6, 18) (c) (14, 28) (d) (15, 30)

10. Find the LCM of each of the following pairs of numbers:
    (a) (2, 3) (b) (4, 6) (c) (7, 4) (d) (5, 7)

11. Draw a chart of numbers from 1 to 100 and use it to write the first five multiples for each of the following numbers:
    (a) 11 (b) 29 (c) 33

12. (a) Write all even numbers between 70 and 83.
    (b) Write all prime numbers between 50 and 99.
    (c) List all odd numbers between 85 and 100.

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Chapter Four

Fractions

Introduction

In this chapter, you will learn about types of fractions, addition and subtraction of fractions with different denominators and mixed numbers.The competencies developed will enable you to apply fractions in real-life situations such as in cooking, science, engineering, time management and other applications.

Think

Life without knowledge of fraction

Types of fractions

There are three main types of fractions which are proper fractions, improper fractions, and mixed numbers

(a) Proper fractions

In a proper fraction, the numerator is less than the denominator. It means, the value of a proper fraction is always less than 1. The following are examples of proper fractions:

$\frac{1}{4}, \frac{2}{4}, \frac{1}{9}, \frac{3}{8}, \frac{10}{12}, \frac{15}{40}$.

(b) Improper fractions

An improper fraction is the opposite of a proper fraction. In an improper fraction, the numerator is greater than or equal to the denominator. The value of an improper fraction is always 1 or

more. The following are examples of an improper fractions:

$\frac{6}{5}, \frac{9}{4}, \frac{10}{7}, \frac{11}{9}, \frac{12}{11}, \frac{4}{3}$.

(c) Mixed numbers

A mixed number is a combination of a whole number and a proper fraction. It is obtained when an improper fraction is simplified. The following are examples of mixed numbers:

$3\frac{1}{2}, 4\frac{1}{2}, 5\frac{3}{4}, 15\frac{1}{4}, 22\frac{1}{3}, 10\frac{1}{3}$.

Fractions with different denominators

Fractions with different denominators are fractions whose bottom numbers are different, for example $\frac{1}{2}$ and $\frac{1}{3}$ have different denominators which are 2 and 3.

Activity 1 — Addition of fractions with different denominators

Steps

1. Use colored paper which is in a circular shape like a pizza, cut it into fraction of $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{6}$ and so on.
2. Add $\frac{1}{2}$ and $\frac{1}{3}$.
3. If $\frac{1}{2}$ and $\frac{1}{3}$ can not be added directly, cut more pieces to make both pizzas into sixths (the least common denominator).
4. Make the equivalent fractions

   (i) show that $\frac{1}{2} = \frac{3}{6}$ and $\frac{1}{3}$ $\frac{2}{6}$ (ii) now add $\frac{3}{6} + \frac{2}{6} = \frac{5}{6}$.

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5. Practice more addition of fraction using fraction cards for

   (i) $\frac{1}{4}+\frac{1}{6}$ (ii) $\frac{2}{3}+\frac{1}{5}$ (iii) $\frac{3}{8}+\frac{1}{4}$

6. Why do we need to find common denominators?

Addition of fractions with different denominators

To add fraction with different denominators, you have to start finding the Lowest Common Multiple (LCM) of the denominators. Then, rewrite each fraction as an equivalent fraction with the LCM as a new denominator. After that, add the numerators while keeping the denimonator the same. Finally simplify the fraction if possible.

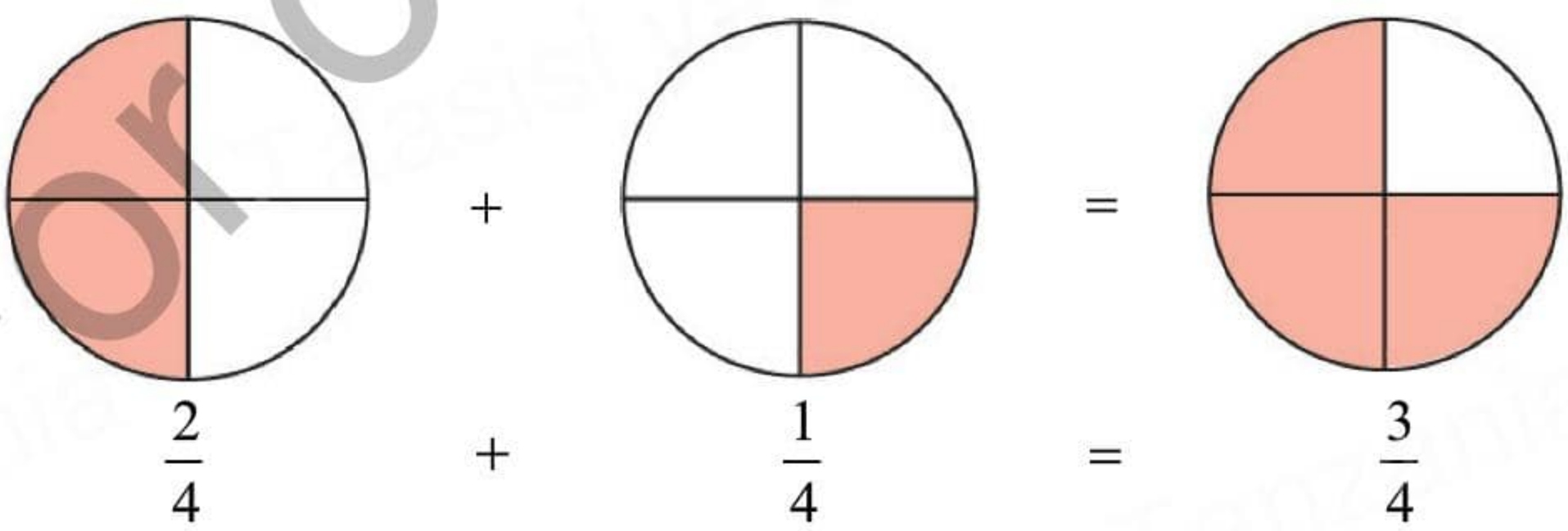
Example 1

$\frac{1}{2}+\frac{1}{4}=$

Solution

Write $\frac{1}{2}$ and $\frac{1}{4}$ as fractions with the same denominator by multiplying $\frac{1}{2}$ by $\frac{2}{2}$ and $\frac{1}{4}$ by $\frac{1}{1}$

That is, $\frac{1}{2}=\frac{1}{2}\times\frac{2}{2}=\frac{2}{4}$.



$\frac{2}{4}$ + $\frac{1}{4}$ = $\frac{3}{4}$

Therefore, $\frac{1}{2}+\frac{1}{4}=\frac{2}{4}+\frac{1}{4}=\frac{2+1}{4}=\frac{3}{4}$.

Example 2

$\frac{2}{5}+\frac{3}{9}=$

Solution

The fractions are of different denominators, change them to have the same denominator:

The Greatest Common Factor of 5 and 9 is 45

Thus, multiply $\frac{2}{5}\times\frac{9}{9}$ and $\frac{3}{9}\times\frac{5}{5}$ to get,

$\frac{2}{5}\times\frac{9}{9}=\frac{18}{45}$ and $\frac{3}{9}\times\frac{5}{5}=\frac{15}{45}$

Thus, $\frac{18}{45}+\frac{15}{45}=\frac{33}{45}=\frac{11}{15}$

Therefore, $\frac{2}{5}+\frac{3}{9}=\frac{11}{15}$.

Example 3

$\frac{4}{15}+\frac{3}{5}+\frac{1}{30}=$

Solution

The given fractions have different denominators, change them to have the same denominators as follows:

$\frac{4}{15}\times\frac{2}{2}=\frac{8}{30}$ and $\frac{3}{5}\times\frac{6}{6}=\frac{18}{30}$

It follows that;

$\frac{4}{15}+\frac{3}{5}+\frac{1}{30}=\frac{8}{30}+\frac{18}{30}+\frac{1}{30}$

$$= \frac{8+18+1}{30} = \frac{27}{30} = \frac{9}{10}$$

Therefore, $\frac{4}{15} + \frac{3}{5} + \frac{1}{30} = \frac{9}{10}$.

Exercise 1

1. $\frac{3}{5} + \frac{1}{6} =$
2. $\frac{3}{4} + \frac{1}{5} =$
3. $\frac{3}{7} + \frac{4}{21} =$
4. $\frac{30}{36} + \frac{1}{18} =$
5. $\frac{1}{5} + \frac{5}{7} =$
6. $\frac{1}{2} + \frac{1}{4} + \frac{1}{5} =$
7. $\frac{5}{20} + \frac{7}{60} =$
8. $\frac{2}{3} + \frac{5}{9} =$
9. $\frac{1}{7} + \frac{2}{5} =$
10. $\frac{8}{20} + \frac{24}{40} =$
11. $\frac{5}{65} + \frac{1}{13} =$
12. $\frac{1}{2} + \frac{1}{3} + \frac{1}{8} =$
13. $\frac{3}{8} + \frac{2}{6} =$
14. $\frac{2}{3} + \frac{1}{9} + \frac{1}{18} =$
15. $\frac{2}{15} + \frac{10}{27} =$
16. $\frac{20}{50} + \frac{3}{25} =$
17. $\frac{3}{4} + \frac{1}{7} + \frac{1}{4} =$
18. $\frac{7}{9} + \frac{10}{27} =$
19. $\frac{12}{25} + \frac{13}{100} =$
20. $\frac{15}{30} + \frac{45}{90} =$

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Activity 2 Subtraction of fractions with different denominators

Steps

1. Use colored paper which is in a circular shape like a pizza cut it into fractions of $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{6}$ etc
2. Subtract $\frac{1}{3}$ from $\frac{1}{2}$
3. If $\frac{1}{2}$ and $\frac{1}{3}$ can not be subtracted directly, cut more pieces to make both pizzas into sixths (the least common denominator)
4. Make the equivalent fractions

   (i) show that $\frac{1}{2}=\frac{3}{6}$ and $\frac{1}{3}=\frac{2}{6}$

   (ii) now subtract $\frac{3}{6}-\frac{2}{6}=\frac{1}{6}$.
5. Practice more subtraction of fraction using fraction cards for:

   (i) $\frac{1}{4}-\frac{1}{6}$ (ii) $\frac{2}{3}-\frac{1}{5}$ (iii) $\frac{3}{8}-\frac{1}{4}$
6. Why do we need to find common denominators?

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Example 1

$\frac{3}{4}-\frac{1}{2}=$

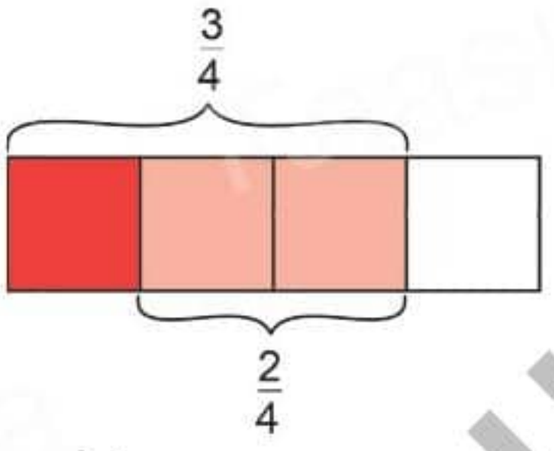
Solution

Multiply $\frac{1}{2}$ by $\frac{2}{2}$ to obtain fractions with the same denominator as follows:

$\frac{1}{2} = \frac{1}{2}\times\frac{2}{2}=\frac{2}{4}$

Thus, $\frac{3}{4}-\frac{1}{2}=\frac{3}{4}-\frac{2}{4}=\frac{1}{4}$.

Alternatively, using rectangular diagram;



equal to

or

$\frac{3}{4}-\frac{1}{2}=\frac{3}{4}-\frac{2}{4}$

$=\frac{3-2}{4}$

$=\frac{1}{4}$

Therefore, $\frac{3}{4}-\frac{1}{2}=\frac{1}{4}$.

Example 2

$\frac{3}{5}-\frac{1}{4}=$

Solution

The LCM of 5 and 4 is 20, then

$\frac{3}{5}-\frac{1}{4}=\frac{(20\div 5)\times 3-(20\div 4)\times 1}{20}$

$=\frac{(4\times 3)-(5\times 1)}{20}$

$=\frac{12-5}{20}$

$=\frac{7}{20}$

Therefore, $\frac{3}{5}-\frac{1}{4}=\frac{7}{20}$.

Example 3

$\frac{5}{6}-\frac{4}{7}=$

Solution

The LCM of 6 and 7 is 42, then

$\frac{5}{6}-\frac{4}{7}=\frac{(42\div 6)\times 5-(42\div 7)\times 4}{42}$

$=\frac{(7\times 5)-(6\times 4)}{42}$

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$$= \frac{35-24}{42}$$

$$= \frac{11}{42}$$

Therefore, $\frac{5}{6} - \frac{4}{7} = \frac{11}{42}$.

Exercise 2

1. $\frac{8}{9} - \frac{2}{6} =$
2. $\frac{4}{8} - \frac{2}{10} =$
3. $\frac{1}{2} - \frac{1}{3} =$
4. $\frac{15}{20} - \frac{14}{40} =$
5. $\frac{25}{30} - \frac{15}{90} =$
6. $\frac{3}{7} - \frac{2}{14} =$
7. $\frac{3}{4} - \frac{4}{9} =$
8. $\frac{6}{8} - \frac{2}{5} =$
9. $\frac{54}{60} - \frac{9}{20} =$
10. $\frac{5}{9} - \frac{2}{4} =$
11. $\frac{8}{15} - \frac{10}{30} =$
12. $\frac{6}{7} - \frac{4}{14} =$
13. $\frac{9}{20} - \frac{3}{10} =$
14. $\frac{8}{12} - \frac{1}{5} =$
15. $\frac{3}{5} - \frac{3}{8} =$
16. $\frac{20}{22} - \frac{8}{11} =$
17. $\frac{24}{28} - \frac{19}{56} =$
18. $\frac{10}{13} - \frac{5}{39} =$
19. $\frac{46}{50} - \frac{40}{100} =$
20. $\frac{18}{30} - \frac{7}{60} =$

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Addition of mixed numbers

Addition of mixed numbers is done by changing mixed number into an improper fraction, then add the improper fraction and finally change back to a mixed number.

Example 1

$2\frac{1}{4}+1\frac{3}{5}=$

Solution

Take the denominator, multiply by the whole number, then add the numerator:

$$2\frac{1}{4}+1\frac{3}{5}=\frac{9}{4}+\frac{8}{5}$$

Find the LCM of 4 and 5 which is 20

Convert fractions with different denominators to a common denominator

$$\frac{9}{4}=\frac{9\times5}{4\times5}=\frac{45}{20}$$

$$\frac{8}{5}=\frac{8\times4}{5\times4}=\frac{32}{20}$$

Add fractions with the same denominator

$$\frac{45}{20}+\frac{32}{20}=\frac{45+32}{20}$$

$$=\frac{77}{20} \text{ or } 3\frac{17}{20}$$

Therefore, $2\frac{1}{4}+1\frac{3}{5}=\frac{77}{20}$ or $3\frac{17}{20}$.

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Alternatively

Steps

1. Add whole numbers; $2+1=3$
2. Add fractions, find the LCM of 4 and 5; $\frac{1}{4}+\frac{3}{5}=\frac{5+12}{20}=\frac{17}{20}$
3. Take 3 obtained in step 1, then add it with the answer obtained in step 2; $3+\frac{17}{20}=3\frac{17}{20}$

Therefore, $2\frac{1}{4}+1\frac{3}{5}=3\frac{17}{20}$.

Example 2

$1\frac{1}{2}+\frac{9}{4}=$

Solution

Take the denominator, multiply by the whole number, then add the numerator: $1\frac{1}{2}+\frac{9}{4}=\frac{3}{2}+\frac{9}{4}$

LCM of 2 and 4 is 4

Convert fractions with different denominators to a common denominator

$\frac{3}{2}=\frac{3\times2}{2\times2}=\frac{6}{4}$

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Add fractions with the same denominator

$$\frac{6}{4}+\frac{9}{4}=\frac{6+9}{4}$$

$$=\frac{15}{4} \text{ or } 3\frac{3}{4}$$

Therefore, $1\frac{1}{2}+\frac{9}{4}=\frac{15}{4}$ or $3\frac{3}{4}$.

Alternatively

Steps

1. Add whole numbers; $1+0=1$
2. Add fractions, find the LCM of 4 and 2: $\frac{1}{2}+\frac{9}{4}=\frac{2+9}{4}=\frac{11}{4}$
3. Take 1 obtained in step 1, then add it to the answer obtained in step 2; $1+\frac{11}{4}=1+2\frac{3}{4}=3\frac{3}{4}$

   Therefore, $1\frac{1}{2}+\frac{9}{4}=3\frac{3}{4}$

Example 3

$5\frac{1}{4}+2\frac{3}{4}=$

Solution

$$5\frac{1}{4}+2\frac{3}{4}=\frac{21}{4}+\frac{11}{4}$$

$$\frac{21}{4}+\frac{11}{4}=\frac{21+11}{4}$$

$$=\frac{32}{4}=8$$

Therefore, $5\frac{1}{4}+2\frac{3}{4}=8$.

Exercise 3

1. $1\frac{1}{3}+2\frac{1}{3}=$
2. $4\frac{2}{5}+3\frac{1}{5}=$
3. $2\frac{3}{4}+1\frac{1}{2}=$
4. $3\frac{12}{20}+\frac{4}{14}=$
5. $1\frac{3}{5}+2\frac{2}{5}=$
6. $5\frac{1}{3}+1\frac{2}{5}=$
7. $3\frac{2}{9}+4\frac{5}{9}=$
8. $2\frac{2}{3}+1\frac{1}{6}=$
9. $1\frac{3}{8}+2\frac{5}{8}=$
10. $3\frac{1}{7}+3\frac{3}{7}=$
11. $3\frac{10}{15}+\frac{10}{30}=$
12. $\frac{8}{9}+7\frac{12}{18}=$

Subtraction of mixed numbers

The subtraction of mixed numbers is done in a similar way as it is done in addition of mixed numbers.

Example 1

$3\frac{2}{5}-1\frac{1}{5}=$

Solution

Take the denominator, multiply it by the whole number, then add the numerator.

$$3\frac{2}{5}-1\frac{1}{5}=\frac{17}{5}-\frac{6}{5}$$

Thus, subtract fractions with the same denominator.

$$\frac{17}{5}-\frac{6}{5}=\frac{17-6}{5}$$

$$=\frac{11}{5}=2\frac{1}{5}$$

Therefore, $3\frac{2}{5}-1\frac{1}{5}=2\frac{1}{5}$.

Alternatively

Steps

1. Subtract whole numbers: $3-1=2$
2. Subtract fractions: $\frac{2}{5}-\frac{1}{5}=\frac{1}{5}$
3. Combine the total, add the whole number obtained in step 1 and the remaining fraction in step 2: $2+\frac{1}{5}=2\frac{1}{5}$

Therefore, $3\frac{2}{5}-1\frac{1}{5}=2\frac{1}{5}$.

Example 2

$$6\frac{1}{6}-2\frac{1}{2}=$$

Solution

Take the denominator, multiply it by the whole number, then add the numerator.

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$$6\frac{1}{6}-2\frac{1}{2}=\frac{37}{6}-\frac{5}{2}$$

The LCM of 2 and 6 is 6.

Convert fractions with different denominators to a common denominator.

$$\frac{5\times3}{2\times3}=\frac{15}{6}$$

Subtract the fractions with the same denominator.

$$\frac{37}{6}-\frac{15}{6}=\frac{37-15}{6}$$

$$=\frac{22}{6}=3\frac{2}{3}$$

Therefore, $6\frac{1}{6}-2\frac{1}{2}=3\frac{2}{3}$.

Alternatively

Steps

1. Subtract whole numbers: $6-2=4$
2. Subtract fractions: $\frac{1}{6}-\frac{1}{2}$, it is not sufficient, take one from the whole number 4 (remain with 3).
3. Convert 1 to $\frac{6}{6}$, add it to the previous fraction: $\frac{1}{6}+\frac{6}{6}=\frac{7}{6}$.
4. Subtract $\frac{7}{6}-\frac{3}{6}=\frac{4}{6}=\frac{2}{3}$
5. Combine the remaining whole number in step 2 and the fraction obtained in step 4: $3+\frac{2}{3}=3\frac{2}{3}$

   Therefore, $6\frac{1}{6}-2\frac{1}{2}=3\frac{2}{3}$.

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Example 3

$7\frac{3}{8}-3\frac{2}{7}=$

Solution

$$7\frac{3}{8}-3\frac{2}{7}=\frac{59}{8}-\frac{23}{7}$$

$$\frac{59\times7}{8\times7}=\frac{413}{56}$$

$$\frac{23\times8}{7\times8}=\frac{184}{56}$$

$$\frac{413}{56}-\frac{184}{56}=\frac{413-184}{56}$$

$$=\frac{229}{56}=4\frac{5}{56}$$

Therefore, $7\frac{3}{8}-3\frac{2}{7}=4\frac{5}{56}$.

Exercise 4

1. $2\frac{2}{5}-\frac{1}{3}=$
2. $6\frac{4}{7}-3\frac{2}{7}=$
3. $4\frac{5}{8}-1\frac{1}{4}=$
4. $8\frac{14}{30}-1\frac{6}{18}=$
5. $2\frac{5}{9}-\frac{4}{13}=$
6. $7\frac{1}{5}-2\frac{4}{9}=$
7. $7\frac{2}{9}-4\frac{5}{9}=$
8. $3\frac{4}{9}-2\frac{1}{9}=$
9. $9\frac{3}{8}-2\frac{5}{8}=$
10. $7\frac{3}{7}-3\frac{1}{7}=$
11. $5\frac{12}{15}-\frac{10}{30}=$
12. $12\frac{8}{18}-7\frac{2}{9}=$

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Activity 3

Learning more about fraction with different denominator and mixed numbers through online lessons

Descriptions

Use various online resources such as Khan Academy to find video lessons and practical exercises on how to work with fractions that have different denominators and mixed numbers to gain a deeper understanding.

Summary

1. When adding or subtracting fractions with different denominators, convert them to have a common denominator, then subtract or add.
2. When adding or subtracting mixed numbers:
   (i) convert the mixed numbers to improper fractions
   (ii) change them to have a common denominator by finding the least common multiple (LCM).
   (iii) subtract or add the fractions with the common denominator.

Revision exercise

1. $8\frac{1}{5}-2\frac{4}{15}=$
2. $\frac{2}{5}-\frac{3}{8}=$
3. $6\frac{7}{12}-3\frac{1}{4}=$
4. $\frac{1}{4}+\frac{5}{6}=$
5. $\frac{11}{12}-\frac{2}{9}=$
6. $\frac{5}{12}+\frac{1}{3}=$

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7. $2\frac{1}{4}+1\frac{3}{8}=$

8. $9\frac{3}{10}-4\frac{7}{10}=$

9. $\frac{7}{9}-\frac{1}{3}=$

10. $10\frac{3}{4}-6\frac{2}{5}=$

11. $\frac{3}{7}+\frac{2}{9}=$

12. $7\frac{5}{8}-3\frac{2}{8}=$

13. $\frac{5}{6}-\frac{1}{4}=$

14. $4\frac{2}{7}+5\frac{5}{14}=$

15. $\frac{7}{10}+\frac{3}{8}=$

16. $12\frac{5}{6}-7\frac{1}{3}=$

17. $3\frac{2}{5}+4\frac{1}{10}=$

18. Ashura used $1\frac{1}{2}$ kg of flour in the morning and $2\frac{1}{4}$ kg in the afternoon. How much flour did she use in total?

19. A jug contains $3\frac{2}{3}$ litres of water. Musa poured out $1\frac{1}{6}$ litres. How many litres of water remained in the jug?

20. Mussa put $4\frac{1}{2}$ litres of milk into a container and he later used $2\frac{2}{5}$ litres. How many litres of milk remained?

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Chapter Five

Decimals

Introduction

In this chapter, you will learn about multiplying and dividing decimals with not more than three decimal places, converting decimals into fractions and fractions into decimals. The competencies developed will enable you to perform various daily life activities such as measuring mass, length, volume and capacity more accurately.

Think

Mathematics without knowledge of decimals

Multiplication of decimals

When multiplying decimals, follow the same procedure as used when multiplying whole numbers without considering the decimal point. Count the required number of decimal places from the right to the left of the product obtained. The number of decimal places in the answer is obtained by counting the total number of decimal places of the multiplied decimal numbers.

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Multiplication of decimals by decimals

Activity 1 Measurements of shirt

Steps

1. Use tape measure to measure across the front of the shirt, from one armpit to the other. Then multiply that number by 2, to get the full chest size.
2. Measure the shoulder width, measure straight across from one shoulder seam to the other.
3. Measure the sleeve length, start at the top of the shoulder seam. Measure all the way down to the end of the sleeve (follow the curve according to the sleeve).
4. Measure the shirt length, measure from the highest point on the shoulder (near the collar) down to the bottom waist.
5. Measure the neck size, measure from the middle button of collar to the end of the buttonhole.
6. Measure the waist, measure straight across the narrowest part of the shirt (usually 6–8 inches below the armpits).
7. Measure the cuff, wrap the tape around the cuff to get the circumference.
8. Record the measurements from step 1 to 7.
9. How much fabric do you need to sew that shirt?

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Multiplication of two decimal places with one decimal place

Example 1

$4.52 \times 1.3 =$

Steps	Solution
1. Multiply these numbers as whole numbers without considering their decimal places 2. That is, multiply 452 by 13 then count three decimal places from right to left of the answer 3. Then, count three decimal places, from right to left to obtain 5.876	$\begin{array}{r} 452 \\ \times\ 13 \\ \hline 452 \\ +1356 \\ \hline 5876 \\ \hline \end{array}$

Therefore, $4.52 \times 1.3 = 5.876$

Remember, when working with decimals, zeros that apper at the end of a decimal after the last non-zero digit do not change the value of a number. Example; 3.400 is the same as 3.4.

Example 2

$100.55 \times 0.2 =$

Steps	Solution
1. Multiply 10055 by 2; then count three decimal places 2. After counting three decimal places, the answer is 20.11	$\begin{array}{r} 10055 \\ \times\ \ 2 \\ \hline 20110 \\ \hline \end{array}$

Therefore, $100.55 \times 0.2 = 20.11$

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Exercise 1

1. $0.11 \times 1.1 =$
2. $0.05 \times 3.5 =$
3. $10.25 \times 4.1 =$
4. $55.22 \times 3.4 =$
5. $66.63 \times 4.7 =$
6. $345.11 \times 0.2 =$
7. $4321.02 \times 0.8 =$
8. $22345.24 \times 0.4 =$
9. $10100.12 \times 0.5 =$
10. $33333.36 \times 0.9 =$
11. $213.21 \times 5.5 =$
12. $196.22 \times 7.1 =$
13. $423.12 \times 2.0 =$
14. $711.30 \times 1.3 =$
15. $32.45 \times 2.3 =$
16. $94.72 \times 4.2 =$
17. $18.62 \times 2.2 =$
18. $111.11 \times 9.3 =$
19. $26.23 \times 1.1 =$
20. The length and width of a rectangle are 12.5 centimetres and 4.6 centimetres, respectively. Find its area.

Multiplying decimals by whole numbers

Example 1

Bakari sells 155.25 litres of water daily, how many litres of water will he sell in 20 days?

Solution

Bakari sells 155.25 litres of water every day.

In 20 days he will sell $155.25 \times 20 = 3{,}105 \text{litres}$

Therefore, Bakari will sell 3,105 litres of water in 20 days

Example 2

$327.425 \times 7 =$

Steps	Solution
1. Multiply 327435 by 7, then indicate three decimal places by counting digits of the answer from the right 2. After indicating three decimal places, the answer is 2292.045	$\begin{array}{r} 327435 \\ \times \quad 7 \\ \hline 2292045 \end{array}$

Therefore, $327.435 \times 7 = 2292.045$

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Example 3

The child's age is 0.25 of the father's age. If the age of the father is 40 years, find the age of the child.

Solution

Multiply 25 by 40, then indicate two decimal places by counting digits of the answer from the right.

$$\begin{array}{r} 25 \\ \times\ 40 \\ \hline 00 \\ +100 \\ \hline 1000 \\ \hline \end{array}$$

After indicating two decimal places, the answer is 10.00

Therefore, the age of the child is 10 years.

Example 4

$27.321 \times 33 =$

Steps	Solution
1. Multiply 27321 by 33, then indicate three decimal places by counting the digits of the answer from the right 2. After indicating three decimal places, the answer is 901.593	$\begin{array}{r} 27321 \\ \times\ \ 33 \\ \hline 81963 \\ +\ 81963 \\ \hline 901593 \\ \hline \end{array}$

Therefore, $27.321 \times 33 = 901.593$

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Exercise 2

1. $50.101 \times 5 =$
2. $1010.014 \times 7 =$
3. $560.222 \times 16 =$
4. $90.32 \times 222 =$
5. $78.545 \times 6 =$
6. $1000.001 \times 9 =$
7. $20.075 \times 20 =$
8. $250.404 \times 15 =$
9. $415.999 \times 24 =$
10. $300.059 \times 37 =$
11. $100.670 \times 5 =$
12. $33.724 \times 9 =$
13. $421.162 \times 2 =$
14. $72.435 \times 7 =$
15. $2000.227 \times 4 =$
16. $360.116 \times 7 =$
17. The speed of an airplane is 732.73 kilometres per hour. How many kilometres will the plane travel in seven hours?
18. Bernad's cow produces 18.5 litres of milk per day. If he has 22 such cows, how many litres of milk will he get in one week?
19. A train wagon carries 16.8 tonnes. If the train has 13 such wagons, how many tonnes will the train carry?
20. Subira and Bahati bought a bicycle for 85,000 shillings. If Subira contributed 0.75 of the buying price, how much did Bahati contribute?

Division of decimals

Division of decimals by whole numbers

To divide a decimal by a whole number, multiply both the numerator and the denominator by the same multiple of 10 depending on the number of decimal places to eliminate decimals.

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Example 1

$0.35 \div 7 =$

Steps	Solution
1. Arrange the two numbers as a fraction	$0.35 \div 7 = \frac{0.35}{7}$
2. Multiply the numerator and the denominator by 10 to remove the decimal places 3. Divide 35 by 700 to get 0.05	$\frac{0.35 \times 100}{7 \times 100} = \frac{35}{700}$ $= 0.05$

Therefore, $0.35 \div 7 = 0.05$

Alternatively

Steps	Solution
1. Divide decimal number as whole number without considering decimal places; 35 by 7 to get 5 2. Separate the decimal places. Count the number of digits after the decimal in the original number, then place the decimal in your answer; 0.05	$\begin{array}{r} 5 \\ 7\overline{)\,35} \\ -35 \\ \hline 0 \end{array}$

Therefore, $0.35 \div 7 = 0.05$

Example 2

A rope of length forty-three point nine four metres was divided into fifty-two equal pieces. How long was each piece in metres?

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Solution

The length of the rope is 43.94 metres.

The number of equal pieces of the rope is 52.

The length of each piece of the rope is obtained as follows:

Divide 43.94 metres by 52 to get the length of each piece.

$$43.94 \div 52 = \frac{43.94}{52} = \frac{43.94 \times 100}{52 \times 100} = \frac{4394}{5200} = 0.845$$

Therefore, each piece of rope had a length of 0.845 metres.

Example 3

$96.75 \div 9 =$

Steps	Solution
1. Arrange the two numbers as a fraction	$96.75 \div 9 = \frac{96.75}{9}$
2. Multiply the numerator and denominator by 100 to remove the decimal places 3. Divide 9675 by 900 to get 10.75	$\frac{96.75 \times 100}{9 \times 100} = \frac{9675}{900}$ $= 10.75$

Therefore, $96.75 \div 9 = 10.75$

Example 4

$45.5 \div 5 =$

Steps	Solution
1. Arrange the numbers as a fraction	$45.5 \div 5 = \frac{45.5}{5}$

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2. Multiply the numerator and denominator by 100 to remove the decimals 3. Divide 455 by 50 to get 9.1	$\frac{45.5\times10}{5\times10}=\frac{455}{50}$ $=9.1$

Therefore, $45.5\div5=9.1$

Exercise 3

1. The total scores of 25 pupils in a Mathematics examination is 1412.5. Find their average score·
2. Standard Six class has 50 pupils. A teacher gave the pupils a rope that measures 8.5 metres. If the pupils shared the rope equally, how long in centimetres did each pupil get?
3. Asha has 25.5 shillings which is to be shared equally among 5 friends. How much money will each friend get?
4. $6.30\div9=$
5. $94.8\div12=$
6. $625.9\div50=$
7. $800.08\div20=$
8. $90.5\div5=$
9. $78.8\div8=$
10. $436.5\div9=$

Division of whole numbers by decimals

Example 1

$24\div0.2=$

Steps	Solution
1. Arrange the numbers as a fraction	$24\div0.2=\frac{24}{0.2}$
2. Multiply the numerator and the denominator by 10 to remove the decimal 3. Divide 240 by 2 to get 120	$\frac{24\times10}{0.2\times10}=\frac{240}{2}$ $=120$

Therefore, $24\div0.2=120$

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Example 2

$65 \div 0.05 =$

Steps	Solution
1. Arrange the numbers as a fraction	$65 \div 0.05 = \frac{65}{0.05}$
2. Multiply the numerator and the denominator by 100 to remove the decimal points 3. Divide 6500 by 5 to get 1300	$\frac{65 \times 100}{0.05 \times 100} = \frac{6500}{5}$ $= 1300$

Therefore, $65 \div 0.05 = 1,300$

Example 3

$125 \div 0.001 =$

Steps	Solution
1. Arrange the numbers as a fraction	$125 \div 0.001 = \frac{125}{0.001}$
2. Multiply the numerator and the denominator by 1000 to remove the decimals 3. Divide 125000 by 1 to get 125000	$\frac{125 \times 1000}{0.001 \times 1000} = \frac{125000}{1}$ $= 125000$

Therefore, $125 \div 0.001 = 125,000$.

Exercise 4

1. $5 \div 0.2 =$
2. $31 \div 15.5 =$
3. $12 \div 0.24 =$
4. $9 \div 0.003 =$
5. $40 \div 8.00 =$
6. $364 \div 0.2 =$

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7. $456 \div 0.6 =$
8. $789 \div 0.05 =$
9. $1250 \div 0.4 =$
10. $450 \div 1.25 =$
11. $5 \div 0.125 =$
12. $77 \div 0.022 =$
13. $969 \div 0.323 =$
14. $17 \div 2.125 =$
15. $616 \div 0.014 =$
16. $848 \div 0.212 =$
17. $3850 \div 0.154 =$
18. $121 \div 0.011 =$
19. $441 \div 0.021 =$
20. $200 \div 3.125 =$
21. $175 \div 0.32 =$

Division of decimals by decimals

Example 1

$5.550 \div 0.111 =$

Steps	Solution
1. Arrange the numbers as a fraction	$5.550 \div 0.111 = \frac{5.550}{0.111}$
2. Multiply both the numerator and the denominator by 1000 to remove the decimals 3. Divide 5550 by 111 to get 50	$\frac{5.550 \times 1000}{0.111 \times 1000} = \frac{5550}{111}$ $= 50$

Therefore, $5.550 \div 0.111 = 50$

Example 2

$52.32 \div 0.05 =$

Steps	Solution
1. Arrange the numbers as a fraction	$52.32 \div 0.05 = \frac{52.32}{0.05}$
2. Multiply both the numerator and the denominator by 100 to remove the decimals 3. Divide 5232 by 5 to get 1046.5	$\frac{52.32 \times 100}{0.05 \times 100} = \frac{5232}{5}$ $= 1046.4$

Therefore, $52.32 \div 0.05 = 1046.4$

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Exercise 5

1. $7.15 \div 0.013 =$
2. $3.8 \div 0.076 =$
3. $4.08 \div 0.2 =$
4. $0.213 \div 0.012 =$
5. $0.919 \div 0.0919 =$
6. $2.147 \div 0.1 =$
7. $6.5 \div 0.05 =$
8. $1.55 \div 0.310 =$
9. $12.5 \div 0.25 =$
10. $19.12 \div 0.2 =$
11. $6.25 \div 0.025 =$
12. $1.9 \div 0.095 =$
13. $0.25 \div 0.125 =$
14. $0.375 \div 0.015 =$
15. $0.909 \div 3.03 =$
16. $2.248 \div 11.24 =$
17. $5.6 \div 0.007 =$
18. $1.7 \div 0.136 =$
19. $11.5 \div 0.092 =$
20. $99.99 \div 0.303 =$

Converting decimals into fractions

To convert decimals into fraction, consider the number of decimal places on a given number. Thus, if the number has one decimal place, its fraction will have a denominator of 10. If the number has two decimal places, its fraction will have a denominator of 100, and if the number has three decimal places, its fraction will have a denominator of 1,000. The numerator will be a whole number after converting the decimal into fraction with a denominator of 10, 100 or 1,000.

Example 1

Convert each of the following decimals into fractions:

(a) 0.2 (b) 0.32

Solution

(a) 0.2

Write the decimal as a fraction: $0.2 = \frac{0.2}{1}$

Multiply both the numerator and the denominator by 10 to remove the decimal.

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$$\frac{0.2\times10}{1\times10}=\frac{2}{10}=\frac{1}{5}$$

Therefore, 0.2 in fraction is $\frac{1}{5}$.

(b) 0.32

Write the decimal as a fraction, that is, $\frac{0.32}{1}$.

Multiply both the numerator and the denominator by 100 to remove the decimals.

$$\frac{0.32}{1}=\frac{0.32\times100}{1\times100}=\frac{32}{100}$$

Simplify $\frac{32}{100}$ to get $\frac{8}{25}$.

Therefore, 0.32 in fraction is $\frac{8}{25}$.

Example 2

Convert the following decimals into fractions:

(a) 0.125 (a) 10.06

Solution

(a) 0.125

Write 0.125 as a fraction, that is, $\frac{0.125}{1}$.

Multiply the numerator and the denominator by 1000. That is,

$$\frac{0.125}{1}=\frac{0.125\times1000}{1\times1000}=\frac{125}{1000}=\frac{1}{8}$$

Therefore, 0.125 in fraction is $\frac{1}{8}$.

(b) 10.06

Write 10.06 as a fraction, that is, $\frac{10.06}{1}$.

Multiply both the numerator and the denominator by 100 to remove the decimals. That is,

$$\frac{10.06}{1}=\frac{10.06\times100}{1\times100}$$
$$=\frac{1006}{100}=\frac{503}{50}$$

In mixed number $\frac{503}{50}$ is equal to $10\frac{3}{50}$.

Therefore, 10.06 in fraction is $\frac{503}{50}$.

Exercise 6

Convert the following decimals into fractions:

1. 0.18	2. 0.154	3. 0.05	4. 0.015
5. 9.564	6. 16.25	7. 50.75	8. 25.125
9. 55.32	10. 96.24	11. 0.375	12. 0.175
13. 40.375	14. 220.08	15. 1000.625	

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Converting fractions into decimals

When converting fractions into decimals, use long division method to divide the numerator by the denominator.

Example 1

Convert $\frac{1}{20}$ into decimal.

Solution

This is a simple fraction. Divide numerator which is 1 by the denominator which is 20 by long division as follows:

Steps	Solution
1. Write 1.00 to allow division with decimals 2. Divide the first number: 1 is not sufficient to divide 20, write 0 and a decimal point in the answer 3. Multiply 0 by 20 to get 0. Write 0 below 1 4. Subtract: $1-0=1$. Bring down the first 0 after the decimal to get 10 5. Divide 10 by 20, it is not sufficient. Write a 0 after the decimal point in the answer 6. Multiply 0 by 20 to get 0. Write 0 below 10 7. Subtract: $10-0=10$. Bring down the second 0 after the decimal to get 100 8. Divide 100 by 20 to get 5 and write 5 in the answer 9. Multiply 5 by 20 to get 100 and subtract $100-100=0$	$20\overline{)1.00}\ 0.05$ -0 10 -0 100 -100 0

Therefore, $\frac{1}{20}=0.05$

Example 2

Convert $\frac{7}{200}$ into decimal.

Solution

This is a simple fraction. Use long division method to divide the numerator 7 by the denominator 200 as follows:

Steps	Solution
1. Write 7.000 to allow division with decimals 2. Divide the first number: 7 is not sufficient to divide 200, write 0 and a decimal point in the answer 3. Multiply 0 by 200 to get 0. Write 0 below 7 4. Subtract: $7-0=7$. Bring down the first 0 after the decimal to get 70 5. Divide 70 by 200, it is not sufficient. Write 0 after the decimal point in the answer 6. Multiply 0 by 200 to get 0. Write 0 below 70 7. Subtruct $70-0=70$. Bring down the second 0 after the decimal to get 700 8. Divide 700 by 200 to get 3 and write 3 in the answer 9. Multiply 3 by 200 to get 600, subtract: $700-600=100$ 10. Bring down the third 0 after the decimal to get 1000 11. Divide 1000 by 200 to get 5 and write 5 in the answer 12. Multiply 5 by 200 to get 1000, subtract: $1000-1000$ to get 0	$200\overline{)7.000}$ -0 70 -0 700 -600 1000 -1000 0

Therefore, $\frac{7}{200}=0.035$

Example 3

Convert $15\frac{3}{4}$ into decimal.

Solution

$15\frac{3}{4}$ is a mixed number of which 15 is a whole number and $\frac{3}{4}$ is a simple fraction. Convert the simple fraction $\frac{3}{4}$ into a decimal by dividing the numerator which is 3 by the denominator which is 4 as follows:

$$
\begin{array}{r}
0.75 \\
4\overline{)\,3.00} \\
-\underline{0} \\
30 \\
-\underline{28} \\
20 \\
-\underline{20} \\
0
\end{array}
$$

That is, $\frac{3}{4} = 0.75$.

Add 0.75 to the whole number. That is $15 + 0.75 = 15.75$

Therefore, $15\frac{3}{4}$ = 15.75

Alternatively

Convert the mixed number into improper fraction, that is, $15\frac{3}{4} = \frac{63}{4}$. Use long division method to divide the numerator, which is 63 by denominator which is 4 as follows:

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$$
\begin{array}{r}
15.75 \\
4\overline{)\,63.00} \\
-4 \\
\hline
23 \\
-20 \\
\hline
30 \\
-28 \\
\hline
20 \\
-20 \\
\hline
0
\end{array}
$$

Therefore, $15\frac{3}{4} = \frac{63}{4} = 15.75$

Example 4

Convert $\frac{41}{2000}$ into decimal.

Solution

This is a simple fraction. Use long division method to divide the numerator 41 by the denominator which is 2000 as follows:

$$
\begin{array}{r}
0.0205 \\
2000\overline{)\,41.0000} \\
-0 \\
\hline
410 \\
-\ 0 \\
\hline
4100 \\
-\ 4000 \\
\hline
1000 \\
-\ \ \ 0 \\
\hline
10000 \\
-10000 \\
\hline
0
\end{array}
$$

Therefore, $\frac{41}{2000} = 0.0205$

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Exercise 7

Convert the following fractions into decimals

1. $\frac{3}{8}$
2. $\frac{3}{4}$
3. $7\frac{1}{8}$
4. $\frac{51}{100}$
5. $\frac{39}{40}$
6. $\frac{13}{50}$
7. $105\frac{3}{4}$
8. $230\frac{1}{20}$
9. $1567\frac{29}{50}$
10. $23954\frac{7}{250}$
11. $345\frac{1}{8}$
12. $36\frac{277}{500}$

Changing percentages into fractions

To convert percentage into fraction, take the percentage value you want to convert and write it as a fraction with denominator of 100. After that simplify the fraction if possible to get the simple form.

Example 1

Change 70% into fraction.

Solution

Remark, $70\% = \frac{70}{100}$

Simplify,

$\frac{70}{100} = \frac{7}{10}$

Therefore, $70\% = \frac{7}{10}$.

Example 2

Change 85% into fraction.

Solution

Remark, $85\% = \frac{85}{100}$

Simplify,

$\frac{85}{100} = \frac{17}{20}$

Therefore, $85\% = \frac{17}{20}$.

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After changing a percentage into a fraction, you can also change a fraction to a percentage by multiplying the fraction by 100.

Exercise 8

1. Change the following percentages into fractions:
   (a) 14% (b) 40% (c) 120% (d) 90% (e) 5%
2. In a certain class, 75% of the pupils attended. What is the fraction of the pupils who attended the class?
3. Linda has fulfilled 40% of her goals. Express the fullfilled goals as a fraction.

Changing percentages into decimals

In order to convert a percentage into decimal, first convert it into a fraction with 100 as the denominator. Then, divide the given number by its denominator to obtain the required decimal.

Example 1

Change 45% into decimals.

Solution

Write 45% as a fraction

$$45\% = \frac{45}{100}$$

Divide to get the decimal

$$45 \div 100 = 0.45$$

Therefore, 45% = 0.45

Example 2

Change 123% into decimal

Solution

Write 123% as a fraction

$$123\% = \frac{123}{100}$$

Divide to get decimal

$$\frac{123}{100} = 1.23$$

Therefore, $123\% = 1.23$

After changing a percentage to a decimal, you can also go back to a percentage by multiplying the decimal by 100, then insert the percentage symbol (%) in the answer.

For example, $1.23 \times 100 = 123\%$.

Exercise 9

1. Change the following percentages into decimals:
   (a) 35% (b) 70% (c) 7% (d) 57% (e) 170%
2. Maria scored 75% in the Mathematics test. Write her score in the decimal.
3. A bottle of water is filled to 60% of its volume. Express the amount as a decimal.
4. A shop is offering a 25% discount on the selling price of shoes. What is this discount as a decimal?

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5. Amani attended school for 90 percent of the first term. Write the percentage in decimal form.
6. John saved 40% of his pocket money, write the percentage as a decimal to show how much he saved.

Activity 2 — Multiplying and dividing decimals through online lessons

Descriptions

Use various programs such as Khan Academy to find video lessons and practical exercises on multiplication and division of decimal numbers for more understanding.

Summary

1. When multiplying decimals, multiply them as whole numbers without considering the decimal points. Thereafter, place a decimal point after counting the decimal places from right to left of the answer. The number of decimal places of the answer is the sum of decimal places of the multiplied decimals.
2. When dividing a decimal by a whole number, write the decimal as a fraction. Remove the decimal points, then divide by using long division method.
3. When dividing decimals by decimals, convert all decimals into whole numbers.
4. When converting fractions into decimals, divide the numerator by the denominator by using the long division method.

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Revision exercise

1. Calculate the following:

   (a) $0.26 \times 12 =$ (b) $157.23 \times 25 =$ (c) $729.126 \times 33 =$

   (d) $0.115 \times 1026 =$ (e) $8 \div 0.5 =$ (f) $147.6 \div 4 =$

   (g) $8345.91 \div 153.7 =$ (h) $123.99 \div 0.03 =$

2. Convert the following decimals into fractions:

   (a) 0.125 (b) 15.01 (c) 560.225 (d) 17.675

3. Convert the following fractions into decimals:

   (a) $\frac{35}{56}$ (b) $460\frac{49}{50}$ (c) $215\frac{39}{40}$ (d) $\frac{132}{165}$

4. Three businessmen bought sugar as follows; 9.5 tonnes, 14.75 tonnes and 23.015 tonnes. How many kilograms of sugar did the businessmen buy in total?

5. A representative of farmers distributed agricultural inputs to 42 farmers. Each farmer received one bag of inputs weighing 12.5 kilograms. How many kilograms did the representative distribute to the farmers?

6. The lengths of two pieces of sugarcane are 84.785 centimetres and 54.7 centimetres, respectively. What is the ratio of the first sugarcane to the second?

7. The area of a rectangle is 1502.55 square centimetres. If the width of the rectangle is 26.5 centimetres, find its length.

8. A crate of soda contains 12 bottles with a mass of 1.5 kilograms each. What is the total mass of the bottles?

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Chapter Six

Algebra

Introduction

In this chapter, you will learn about the concept of algebra, identify terms, variables and coefficients in algebraic expressions. You will also learn how to use arithmetic operations to express algebraic expressions and their simplifications. The competencies developed will enable you to transform real-life problems into mathematical statements and solve them effectively.

Think

Mathematics without a knowledge of algebra

The concept of algebra

Algebra is a branch of mathematics in which numbers and letters are used. These letters are used to represent numbers or unknown quantities. Algebra uses letters to solve real-life problems and mathematical equations, for example performing calculations when the value of the letter is unknown.

Algebraic expressions

Algebra is used in everyday life to simplify mathematical calculations. For example; 4 families rear chicken and each family has the same number of chicken. If k represent the number of chicken in one family, then, the total number of chickens owned by the four families is: $k+k+k+k=4k$

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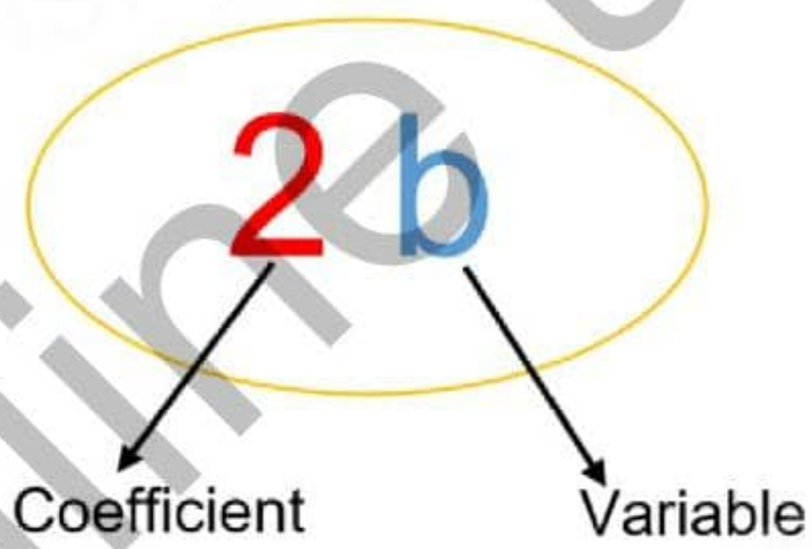
Similarly, suppose that, the four families keep goats as follows: the first family has twice as many goats as chicken, the second family has thrice as many goats as chicken, the third family has twice as many goats as chicken and the fourth family has four times as many goats as chicken. The total number of goats owned by the four families can be written as:

$$2k + 3k + 2k + 4k = 11k.$$

In this example; $2k$, $3k$, $4k$ and $11k$ are called terms.

In the term $4k$, the number 4 is called the **coefficient** of k and the alphabet k is called the **variable**. Similarly, in the term $11k$, 11 is the **coefficient** of k and k is the **variable**. A term may also have two or more variables, such as in the expression $8pq$, where 8 is the coefficient of p and q, while p and q are **variables**



Example 1

Identify the coefficient and the variable in each of the following terms:

(a) $6a$ (b) y

Answer

(a) In the term $6a$, 6 is the coefficient and a is the variable.

(b) In the term y, 1 is the coefficient and y is the variable.

Example 2

Write the variable and coefficient in each of the following terms:

(a) $3pq$ (b) $5wz$

Answer

(a) In the term $3pq$, p and q are variables 3 is the coefficient.

(b) In the term $5wz$, w and z are variables and 5 is the coefficient of w and z.

Algebraic expressions are constructed using terms such as $4k$, $11m$ and $12w$ connected with arithmetic operations. The following are examples of algebraic expressions:
$4k + 6k$, $11m - m$, $6m \times 3$, $12w \div 2n$.

Exercise 1

1. Write the variables in each of the following terms.

   (a) $7mn$ (b) $5p$ (c) n (d) $10tp$ (e) $3yw$

2. Write the coefficient in each of the following terms.

   (a) $3pq$ (b) $4ab$ (c) pk (d) m (e) a

Addition and subtraction of algebraic terms

Terms in the algebraic expression are said to be alike if they have the same variable, such as $4k, 8k, 5k$ and $7k$. Like terms can be added or substracted to obtain a simpler algebraic term. In adding like terms, you add the coefficients and leave the variable unchanged. Terms which have different variables are called unlike terms. For example; $3a, 4m, 6p$ and $7w$. Unlike terms can not be added or substracted.

Addition and subtraction of operations may be combined to simplify a long algebraic expression.

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Example 1

Simplify $6n + n + 2n$

Solution

First, add the coefficients: $6 + 1 + 2 = 9$

Therefore, $6n + n + 2n = 9n$.

Example 2

Simplify $5ae - 2ab - 2ae$

Solution

Collect like terms and then simplify

$$5ae - 2ae - 2ab = (5 - 2)\ ae - 2ab$$
$$= 3ae - 2ab$$

Therefore, $5ae - 2ae - 2ab = 3ae - 2ab$.

Example 3

Simplify $2a + 3b - a$

Solution

Collect like terms and simplify

$$2a + 3b - a = 2a - a + 3b$$
$$= a + 3b$$

Therefore, $2a + 3b - a = a + 3b$

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Example 4

Simplify $4xy - xw + 3xw + 3xy$

Solution

Collect like terms and simplify

$$\begin{aligned} 4xy - xw + 3xw + 3xy &= 4xy + 3xy + 3xw - xw \\ &= (4 + 3)xy + (3 - 1)xw \\ &= 7xy + 2xw \end{aligned}$$

Therefore, $4xy - xw + 3xw + 3xy = 7xy + 2xw$.

Example 5

Simplify $3a - a + 4pq - pq$

Solution

Collect like terms and simplify

$$\begin{aligned} 3a - a + 4pq - pq &= 3a - a + 4pq - pq \\ &= (3-1)a + (4-1)pq \\ &= 2a + 3pq \end{aligned}$$

Therefore, $3a - a + 4pq - pq = 2a + 3pq$.

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Exercise 2

Simplify the following expressions:

1. $3p + 4p + 5p$
2. $2p + q + b + 3b + 2q$
3. $6gh + 6gh + 6gh$
4. $x + 4w + 2w + 2w + x + 3x$
5. $6t - t - 2t$
6. $8xy + xy + 5xy$
7. $7mk - 4mk - mk$
8. $8ad - ad - 2ad$
9. $15k - 4k + 3k - k$
10. $16xy - 14xy + 8$
11. $10b + 2a - 5b$
12. $16pa + 4q - 6pa - 3q$
13. $16y - 10y + 2a + 11y$
14. $8wz + 8wy - wz - 2wy$
15. $6e - 4f + 6f - e$
16. $4w + 6z - 2w - 2z$
17. $4n + 2n + n$
18. $9m + 6n - m + n$
19. $10a - 10b - 10a + 10b$
20. $xy + 3xy + 2xy - xy$

Multiplication of algebraic terms

Algebraic terms can be multiplied by a number or by another like or unlike terms. In multiplying algebraic terms, multiply the coefficients and then multiply the variables. In multiplying terms by a number, multiply the coefficient by the number and then multiply the variables. If the number is zero, then the product is 0.

Example 1

$y \times y$

Solution

In this expression, y is the only variable involved, and both coefficients are 1. When a variable is multiplied by itself, the product is its square.

Therefore, $y \times y = y^2$

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Example 2

Simplify $4a \times 4$

Solution

In this question, an expression is multiplied by 4.

Therefore, $4 \times 4a = 16a$.

Example 3

Simplify $3k \times 2k$

Solution

This expression is a product of two like terms. The variable is k and the coefficients are 3 and 2.

Multiply the coefficients and variables: $3k \times 2k = (3 \times 2) \times (k \times k)$

$= 6k^2$ (6 is the coefficient and k is the variable)

Therefore, $3k \times 2k = 6k^2$.

Exercise 3

Simplify the following expressions:

1. $b \times b$
2. $2n \times 2n$
3. $3p \times p$
4. $3y \times 3y$
5. $t \times k \times r$
6. $4a \times 3b \times 2c$
7. $3a \times 3b \times a$
8. $m \times m$
9. $4k \times 4t$
10. $3a \times 2e \times 4$
11. $2p \times t \times 3p$
12. $y \times w \times 2y \times 3w$

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13. $4a \times b \times a \times b$

14. $10t \times 10p$

15. $6p \times q \times 6p$

16. $3b \times 3 \times 4b$

17. $3na \times 4 \times 2n$

18. $p \times q \times 5q \times 5w$

19. $3tp \times 0$

20. $7vfk \times kp$

Division of algebraic terms

In dividing algebraic terms, divide the coefficients and the like variables.

Example 1

Simplify $4a \div 2a$

Solution

Identify the coefficients and the variables and then simplify.

$4a \div 2a = (4 \div 2) \times (a \div a)$.

Divide coefficient by coefficient: $4 \div 2 = 2$

Then, divide variable by variable: $a \div a = 1$

Multiply the answer of coefficient and variable $= 2 \times 1 = 2$

Therefore, $4a \div 2a = 2$.

Example 2

Simplify $4b \div 4b$

Solution

Identify the coefficients and the variables:

$4b \div 4b = (4 \div 4) \times (b \div b)$

Divide the coefficients $4 \div 4 = 1$. Then, divide the variables:

$b \div b = 1$.

Therefore, $4b \div 4b = 1$.

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Example 3

Simplify $4k^2 \div 2k$

Solution

$$4k^2 \div 2k = (4 \div 2) \times (k^2 \div k)$$
$$= 2 \times k$$
$$= 2k$$

Therefore, $4k^2 \div 2k = 2k$

Exercise 4

Simplify the following expressions:

1. $6 \div 2$
2. $3b \div b$
3. $4c \div 4c$
4. $8b \div 2b$
5. $6b \div 3b$
6. $10y \div y$
7. $15a \div 5a$
8. $5a^2 \div a$
9. $8k^2 \div 8$
10. $10p^2 \div 2p$
11. $t^2 \div t$
12. $4m^2 \div 2m^2$
13. $6k^2p \div 2p$
14. $10n^2 \div 10n$
15. $18x^2 \div 9x$
16. $10y^2p \div 10yp$
17. $12nm^2 \div 12m$
18. $10a^2b^2 \div 10ab$
19. $12m^2n \div 3m^2$
20. $8y^2w^2 \div 4yw^2$

Equations

The word “equation” is derived from the verb “equal”. An equation is like a beam balance whereby two sides must be balanced. An equation is a mathematical statement that show two expressions connected by an equal sign.

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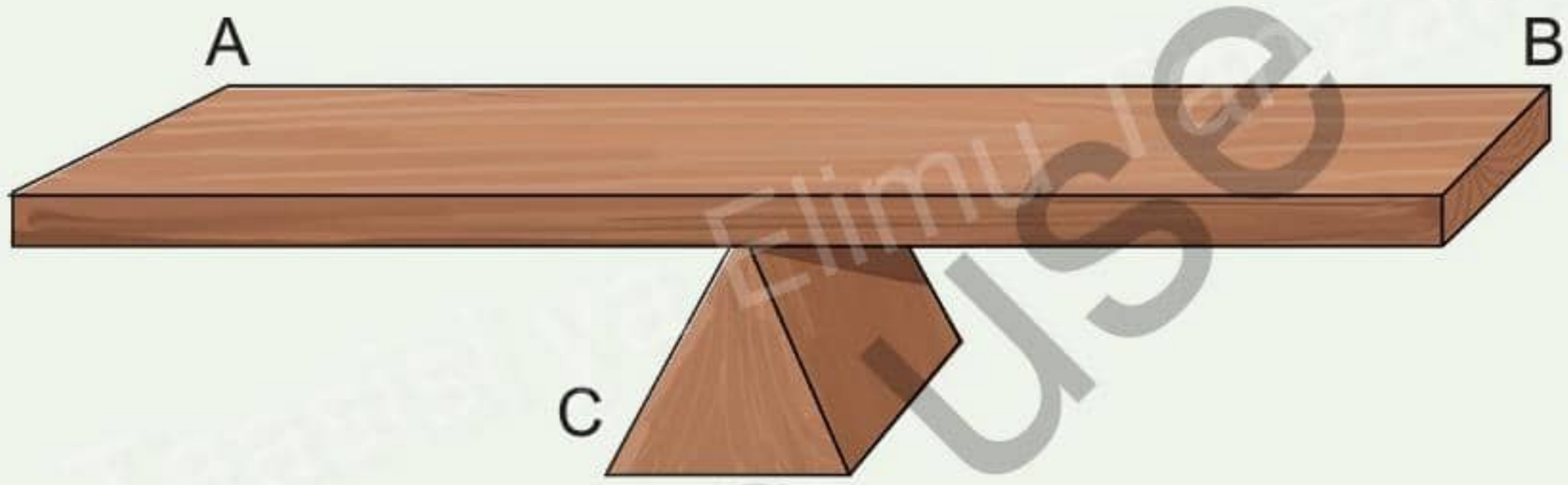
There are simple equations and not simple equations. Simple equations have one variable whose power is one, whereas, not simple equations have more than one variables and have power larger than one.

Activity 1 Explaining the concept of equations

Use a piece of wood or timber to make the following instrument, then use it to perform the balancing game.



Steps

1. Put two stones A and B, or any other objects of the same size, in two sides.
2. The object labeled C is placed between A and B as shown in the figure.
3. If the stones A and B are placed on one side of the balance and the equation tilts to one side, it means that it is not an equation, because the mass on the one side will be greater than the mass on the other side.
4. For the equation to balance, the stones A and B should be of the same mass.

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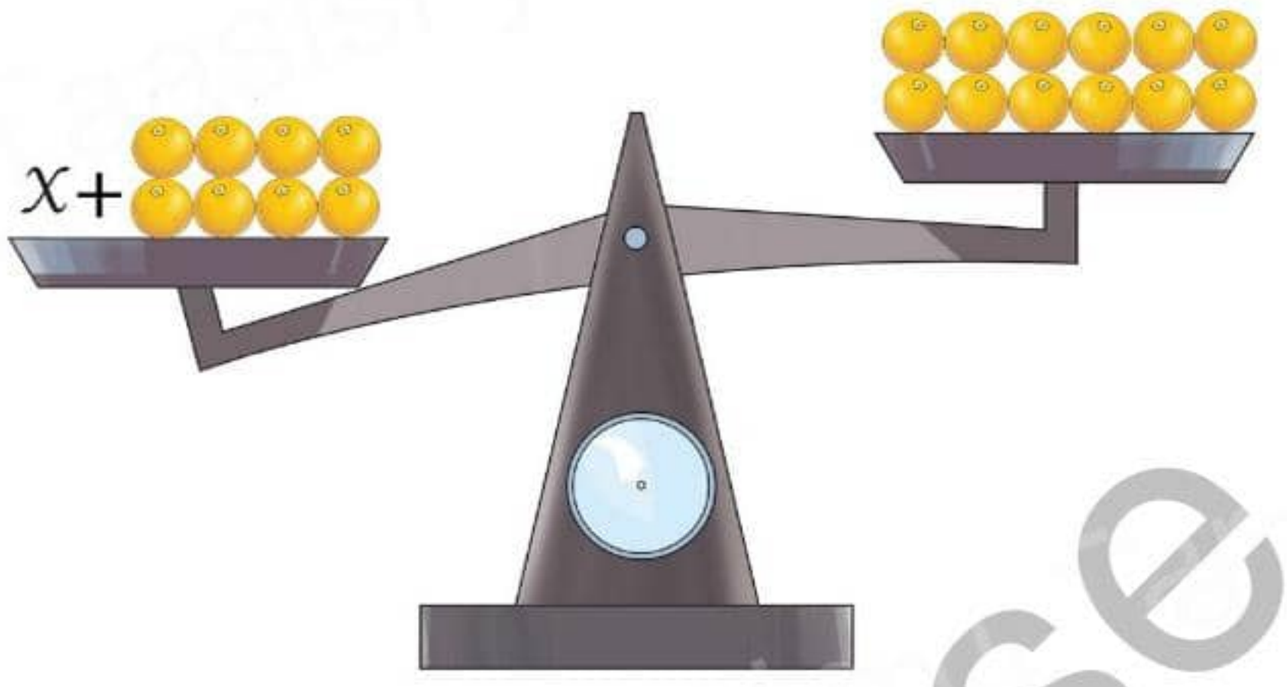
Example 1

Letter x represents the number of oranges required to make the two sides of the beam balance equal. How many oranges are represented by x, if they have the same mass.



For the two sides of the beam balance to be equal, four more oranges are required on the left-hand side. Therefore, algebraic expressions that consider equal values on the right and left sides using a variable are called equations.

Example 2

Study the following equations and identify simple equations:

(a) $3x+5=16$

(b) $2xy-7=2x+3$

(c) $x^2-16=0$

(d) $4(y+5)=3(y-1)$

Answers

Equations (a) and (d) are simple equations because each has one variable and its highest exponent is one.

Equation (b) is not a simple equation because the first term on the left side, $2xy$ has two variables x and y.

Equation (c) is not a simple equation because the highest exponent of x is 2.

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Exercise 5

Which of the following are simple equations?

1. $x-3=3$
2. $a^2-1=10$
3. $9(y+5)=4(y-1)$
4. $x-37=2x-3$
5. $3x^2y-1=16$
6. $y+5=0$
7. $2c-7c=2a+3$
8. $k-16=3k-3$
9. $4=3(y-1)$
10. $w+5=3(w-1)$

Solving simple equations

When solving simple equations, collect terms with variables to the one side of the equation and numbers to the other side of the equation. Adding or subtracting same numbers on both sides of the equation does not change the meaning of the equation. Multiplying or dividing by the same numbers on both sides of the equation does not change the meaning of the equation.

Addition in simple equations

Example 1

Find the value of p in the following equation: $p+2=3$.

Solution

To make p the subject of the equation

Subtract 2 from both sides:

$p+2-2=3-2$

$p=1$

Therefore, the value of p is 1.

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Example 2

Find the value of y if, $y+12+2y=69$

Solution

Combine the like terms, then add

$$y+2y+12=69$$

$$3y+12=69$$

Subtract 12 from both sides of the equation

$$3y+12-12=69-12$$

$$3y+0=57$$

$$3y=57$$

Divide by 3 both sides of the equation.

$$\frac{3y}{3}=\frac{57}{3}$$

$$y=19$$

Therefore, the value of y is 19.

Example 3

Rehema is 5 years older than Kemi. If the sum of their ages is 25 years. Find their ages.

Solution

Form an equation by choosing any variable.

Let x be the age of Kemi.

Note that; Rehema is 5 years older than Kemi. Thus, add 5 to Kemi's age to get Rehema's age.

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That is, Rehema's age is $x+5$

So the simple equation is, $(x+5)+x=25$

Collect the terms with variable, then add them

$$x+x+5=25$$
$$2x+5=25$$

Subtract 5 from both sides of the equation

$$2x+5-5=25-5$$
$$2x+0=20$$
$$2x=20$$

Divide both sides by 2 to get the value of x

$$\frac{2x}{2}=\frac{20}{2}$$
$$x=10$$

Thus, Rehema's age is $5+10=15$

Therefore, Kemi's age is 10 years and Rehema's age is 15 years.

Example 4

The age of Furaha is $\frac{1}{2}$ the age of her aunt. If the sum of their ages is 54 years, find their ages.

Solution

Let x be the age of Furaha's aunt

The age of Furaha is $\frac{1}{2}x$

But the sum of their ages is 54 years

Thus,

$$x+\frac{1}{2}x=54$$

$$\frac{2x+x}{2}=54$$

$$\frac{3x}{2}=54$$

Multiply both sides of the equation by 2 to get:

$$\frac{3x}{2}\times 2=54\times 2$$

$$3x=108$$

Divide by 3 on both sides:

$$\frac{3x}{3}=\frac{108}{3}$$

$$x=36$$

Thus, the age of Furaha's aunt is 36 years. The age of Furaha is given by:

$$\frac{1}{2}x=\frac{1}{2}\times 36=18$$

Therefore, the age of Furaha is 18 years.

Exercise 6

Find the value of each variable in the following equations:

1. $t+2=12$
2. $a+13=43$
3. $9=4+n$
4. $7=y+3$
5. $x+5=15$
6. $m+8=9$

7. $n+2n-11=4$

8. $a-8+3a=4$

9. $60y+40+59y=170$

10. $21=6p+8p+7$

11. $0.2x+22=88$

12. $5.5n+55+10.5n=311$

13. $6.25t+0.05=6.3$

14. $\frac{4}{5}w+\frac{1}{5}=\frac{21}{5}$

15. $\frac{1}{3}+\frac{1}{3}c=\frac{6}{7}$

16. $\frac{9}{s}+\frac{2}{5}=\frac{7}{4}$

17. $\frac{x}{3}+\frac{1}{5}+\frac{4x}{3}=\frac{16}{5}$

18. $775000=55000r+225000$

19. $400000m+800000=1200000$

20. $\frac{250000}{5}p+\frac{100000}{7}=\frac{450000}{7}$

21. Think of a number, multiply it by 3 and add 5 to the product. If the answer is 20, what is the number?

22. Think of a number, multiply it by 8, and add 15 to the product. If the answer is 22, what is the number?

Subtraction in simple equations

Example 1

Find the value of k in the equation $k-5=7$.

Solution

To make k remain alone on the left-hand side of the equation, add 5 on both sides.

$k-5+5=7+5$

$k+0=12$

$k=12$

Therefore, the value of k is 12.

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Example 2

Ali bought a bus ticket for 450 shillings to travel to school and remained with 1,750 shillings. How much money did he have?

Solution

Let y be the total amount of money Ali had before buying a ticket. Therefore, the equation will be:

$$y-450=1750$$

Add 450 on both sides of the equation:

$$y-450+450=1750+450$$

$$y+0=2200$$

$$y=2200$$

Therefore, Ali had 2,200 shillings.

Example 3

Find the value of v in the equation, $v-19=14$.

Solution

To make v remain alone in the left hand side, add 19 in both sides of the equation:

$$v-19+19=14+19$$

$$v=33$$

Therefore, the value of v is 33.

Exercise 7

Find the value of the variable in each of the following equations:

1. $c-7=2$
2. $f-14=12$
3. $w-3=12$
4. $y-3=1$
5. $x-48=2$
6. $q-0.3=0.2$
7. $n-40=20$
8. $7a-21-6a=10$
9. $14y-5-13y=30$
10. $a-\frac{1}{2}=13$
11. $0.4x-4.4=17.6$
12. $31.5p-165-16.5p=1335$
13. $0.5a-1=6.50$
14. $\frac{8}{5}x-\frac{2}{5}=\frac{46}{5}$
15. $\frac{9}{3}-\frac{5}{3}z=\frac{4}{3}$
16. $\frac{9}{c}-\frac{2}{5}=\frac{1}{2}$
17. $\frac{19}{5}=\frac{a}{3}-\frac{1}{5}+\frac{2a}{3}$
18. $775=25k-225$
19. $500m-800=200$
20. $19.5-7x=12.5x$
21. Joseph bought a pen at 200 shillings and remained with 2,300 shillings. How much money did he have before buying the pen?
22. Amina sold 800 trees from her farm and 324 trees remained in the farm. How many trees were there in the farm?
23. Think of a number and subtract $\frac{1}{8}$ from it. If the answer is 41, what number is that?

Multiplication in simple equations

The opposite of multiplication is division. Therefore, in solving equations which involve multiplication, you can also apply division.

Example 1

Find the value of n if $4 \times n = 12$.

Solution

$$4 \times n = 12$$
$$4n = 12$$

Divide by 4 on both sides of the equation to get the value of n.

$$\frac{4n}{4} = \frac{12}{4}$$
$$n = 3$$

Therefore, the value of n is 3.

Example 2

Find the value of k if $\frac{3}{4}k \times \frac{1}{3} = 24$.

Solution

Multiply terms on the left-hand side of the equal sign (=). That is, numerator × numerator and denominator × denominator.

$$\frac{3}{4}k \times \frac{1}{3} = \frac{3}{12}k$$

Thus, $\frac{3}{12}k = 24$.

In order to convert the fraction into whole number, multiply both sides by 12:

$$12 \times \frac{3}{12}k = 24 \times 12$$
$$3k = 288$$

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To make k remain alone in the left hand side, divide both sides of the equation by 3

$$\frac{3k}{3}=\frac{288}{3}$$

$$k=96$$

Therefore, the value of k is 96.

Exercise 8

Find the value of the variable in each of the following equations:

1. $x\times 4=36$
2. $7=3y$
3. $4w\times 20=8$
4. $3=81k$
5. $6v\times\frac{1}{5}=\frac{12}{5}$
6. $\frac{1}{2}r\times\frac{8}{6}=\frac{9}{4}$
7. $1\frac{3}{7}=\frac{20}{7v}$
8. $0.3p\times 0.1=0.01$
9. $0.5t\times 0.25=500$
10. $100=n\times 0.5$
11. $\frac{2}{3p}\times\frac{2}{3}=\frac{4}{9}$
12. $300y\times\frac{2}{3}=900$
13. $\frac{2}{3z}\times 450=900$
14. $\frac{k}{3}\times 240=720$
15. A school has 249 girls. If the number of girls is $\frac{3}{4}$ of all the pupils, how many pupils are at the school?
16. The area of a rectangle is $30\,\text{m}^2$. If its width is p metres and its length is 8 metres, find the width of the rectangle.
17. The height of a triangle is k cm. If its area is $30\,\text{cm}^2$ and its base is 6 cm, what is the value of k?
18. The number of pupils who attended school is 0.8 of all the pupils. If 720 pupils attended to school, what is the total number of pupils at the school?

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19. The price of a chicken is 8,000 shillings. How many chickens will be sold to get 72,000 shillings.

20. If $5n+15$ children ate 350 oranges, find the value of n if every child ate one orange.

Division in simple equations

The opposite of division is multiplication. Therefore, when solving simple equations which involve division you have to multiply both sides of the equation by the denominator.

Example 1

Find the value of x if $x \div 2 = 4$.

Solution

$x \div 2 = 4$

Write the equation in fraction to get the numerator and denominator:

$\frac{x}{2} = 4$

Multiply both sides of the equation by 2 (the denominator)

$2 \times \frac{x}{2} = 4 \times 2$

$x = 8$

Therefore, the value of x is 8.

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Example 2

Find the value of m if, $2 \div m = 7$.

Solution

$2 \div m = 7$ can be written as $\frac{2}{m} = 7$

Multiply both sides of the equation by m (the denominator)

$$m \times \frac{2}{m} = 7 \times m$$

$$2 = 7m$$

Divide both sides of the equation by 7 to get the value of m.

$$\frac{2}{7} = \frac{7m}{7}$$

$$\frac{2}{7} = m$$

Therefore, the value of m is $\frac{2}{7}$.

Exercise 9

Find the value of the variable in each of the following equations:

1. $x \div 4 = 5$
2. $3 = 2y \div 1$
3. $4w \div 20 = 8$
4. $k \div 3 = 81$
5. $625 \div \frac{t}{5} = 125$
6. $\frac{1}{2}r \div \frac{2}{8} = \frac{1}{4}$
7. $\frac{3}{4} = \frac{1}{5}v \div \frac{2}{5}$
8. $0.3x \div 0.1 = 4$
9. $2.5t \div 1.25 = 500$
10. $\frac{2}{3p} \div \frac{2}{3} = 1$

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Solving simple equations with a variable on both sides

Example 1

Find the value of x if $7x-6=x+18$.

Solution

$7x-6=x+18$

Collect terms with variables on one side and the numbers on the other side:

$7x-x=18+6$

Simplify terms with variables and those without variables:

$6x=24$

Divide both sides of the equation by 6

$$\frac{6x}{6}=\frac{24}{6}$$

$x=4$

Therefore, the value of x is 4.

Example 2

Find the value of b if $3b-4=b+8$

Solution

Collect the terms with variable on one side and numbers on the other side:

$3b-b=8+4$

Simplify the like terms

From $3b-b=8+4$

$2b=12$

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Divide both sides of the equation by 2:

$$\frac{2b}{2}=\frac{12}{2}$$

$$b=6$$

Therefore, the value of b is 6.

Exercise 10

Find the value of the variable in each of the following equations:

1. $3x+5=29-5x$
2. $3y+1=y+7$
3. $w+3=4-2w$
4. $7(k-1)=9-k$
5. $22r+4=52+14r$
6. $7c+2=5c+18$
7. $12y-22=y$
8. $20r-7=2r+23$
9. $3(t-1)=2t+1$
10. $\frac{1}{2}r-1=\frac{1}{4}r+2$
11. $\frac{3}{4}(v-1)=\frac{1}{5}v\div\frac{2}{5}$
12. $0.3x\div0.1=0.9$
13. $2.5t-1.25=0.5t+2.75$
14. $10-6b=7b-29$

Activity 2 **Algebra through online lessons**

Descriptions

Use various programs such as Khan Academy to find video lessons and practical exercises to deepen understatnding on Algebra.

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Summary

1. Algebraic expressions are constructed using the terms.
2. Algebraic terms are made up of coefficients and variables.
3. Like terms have the same variables.
4. To add like terms, add the coefficients. The variable remains the same.
5. To subtract like terms, subtract the coefficients. The variable remains the same.
6. Unlike terms cannot be combined to obtain a single term, but they may be simplified.
7. To multiply algebraic terms, multiply the coefficients, as well as the variables.
8. A simple equation is an arrangement of algebraic terms which require the left side and the right side to be equal.
9. A simple equation has one variable whose power is 1.
10. When collecting like terms to the other side of an equal sign, the operation sign changes.
11. Adding or subtracting the same number on both sides of the simple equation does not change the meaning of the equation.
12. Multiplying or dividing the same number on both sides of the simple equation does not change the meaning of the equation.

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Revision exercise

Simplify the expressions in question 1 - 9 and find the value of the variable in queation 10 -16.

1. $3n+n$
2. $8k-4k$
3. $3a\times b\times b$
4. $16m\div 4m$
5. $10+2y-10=$
6. $5a+3b-2a-2b$
7. $6y^2+2x^2-2y^2+3x^2$
8. $10ab^2c\div 5ab$
9. $4k^2\div 2k^2$
10. $y+2y+5y=\frac{1}{2}$
11. $0.03x+0.1=0.12$
12. $\frac{4}{5}y=4-\frac{1}{5}y$
13. $\frac{3}{4}c-\frac{2}{4}c=\frac{1}{16}-\frac{1}{8}c$
14. $5a-\frac{9}{2}=65-\frac{a}{2}$
13. $0.5f\times 0.25=0.0625$
16. $\frac{3}{4}\div j=1\frac{1}{2}$
17. Think of a number, multiply it by $\frac{3}{4}$, and subtract 6 from the product. If the answer is 66, what is the number?
18. Think of a number, divide it by 7 and add 6 to the result. If the answer is 3 times the number, find the value of this number.
19. George is 3 years older than Abduli. If the sum of their ages is 37, find the age of George?
20. Juma is 5 years older than Anna. If the sum of their ages is twenty-five years, find their ages.

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Chapter Seven

Financial mathematics

Introduction

In this chapter, you will learn about solving problems on sales and purchases in business. Additionally, you will learn to prepare invoices of quantities. The competencies developed will enable you to manage business activities effectively and create awareness of profit and loss in business.

Think

Doing business without having the knowledge of financial mathematics

Profit and loss

Profit is an excess amount of money obtained after selling goods. Profit is obtained if the selling price is greater than the buying price. Thus, to obtain the profit the following formula is used:

$$\text{Profit} = \text{Selling price} - \text{Buying price}$$

Loss is the opposite of profit. Loss is obtained when the selling price is less than the buying price. Therefore, loss is the difference between the buying and selling price. That is;

$$\text{Loss} = \text{Buying price} - \text{Selling price}$$

Word problems involving sales and purchases

Example 1

An orange was bought at 150 shillings and sold at 200 shillings:

(a) what was the profit generated?

(b) what was the percentage profit?

Solution

(a) Profit is the difference between selling price and buying price.

Selling price = shs 200

Buying price = shs 150

Profit = Selling price − Buying price

Profit = shs 200 − shs 150 = shs 50

Therefore, the generated profit was shs 50.

(b) Percentage profit is obtained by expressing the profit as a fraction of the buying price:

$$\frac{50}{150} = \frac{1}{3}$$

Multiply the fraction by 100 to obtain the percentage profit, that is: $\frac{1}{3} \times 100 = 33.3\%$

Therefore, the profit was 33.3%.

Example 2

Juma bought a pair of shoes at 25,000 shillings and sold it at 16,000 shillings:

(a) what loss did Juma get?

(b) what was the percentage loss?

Solution

(a) Loss is the difference of buying price and selling price.

$$\text{Selling price} = \text{shs } 16{,}000$$

$$\text{Buying price} = \text{shs } 25{,}000$$

$$\text{Loss} = \text{Buying price} - \text{Selling price}$$

$$\text{Loss} = \text{shs } 25000 - \text{shs } 16000 = \text{shs } 9{,}000$$

Therefore, Juma got a loss of 9,000 shillings.

(b) Percentage loss is obtained by expressing the loss as a fraction of buying price.

$$\frac{9000}{25000} = \frac{9}{25}$$

Multiply the fraction representing the loss by 100, to obtain the percentage:

$$\frac{9}{25} \times 100 = 36\%$$

Therefore, the loss was 36%.

Example 3

The carpenter made a door at a cost of 250,000 shillings and sold it at a profit of 10%:

(a) what profit did the carpenter get?

(b) how much money did the carpenter get in total?

Solution

(a) The cost of making a door $= \text{shs } 250{,}000$.

To obtain the profit, calculate 10% of shs 250,000 as follows:

$$10\% = \frac{10}{100}$$

Multiply the percentage profit by the cost of making a door, that is

$$\frac{10}{100} \times 250000 = 25,000$$

Therefore, the carpenter got a profit of 25,000 shillings.

(b) The total amount of money that the carpenter got, is the sum of money for making the door and the profit made:

$$\text{shs } 250000 + \text{shs } 25000 = \text{shs } 275,000$$

Therefore, the carpenter got a total of 275,000 shillings.

Example 4

Amani bought 40 goats at shs 3,200,000 and sold them at the open market at a loss of 18%:

(a) what loss did Amani get in shillings?

(b) how much money did he get from the sales?

Solution

(a) To obtain the loss, calculate 18% of 3,200,000 shillings as follows:

$$18\% = \frac{18}{100}$$

$$\frac{18}{100} \times 3200000 = 576,000$$

Therefore, Amani got a loss of 576,000 shillings.

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(b) The amount of money that Amani got is the difference between the buying price and the loss, that is

shs 3200000 – shs 576000 = shs 2,624,000

Therefore, the amount of money Amani got after the sale was 2,624,000 shillings.

Example 5

The selling price of the commodity is 2,100,000. Find its buying price if it is sold at a 20% loss.

Solution

Let x be the buying price

Thus, loss $20\% \times x = \frac{20}{100} x = 0.2x$

But,

$$\text{Loss} = \text{Buying price} - \text{Selling price}$$

$$0.2x = x - 2100000$$

$$x - 0.2x = 2100000$$

$$0.8x = 2100000$$

$$x = \frac{2100000}{0.8}$$

$$x = 2,625,000$$

Therefore, the buying price is $2,625,000$ shillings.

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Exercise 1

1. A gas cooker was bought at 180,000 shillings and sold at 225,000 shillings:
   (a) what was the profit?
   (b) what was the percentage profit?

2. If Halima would have sold her cupboard for 1,200,000 shillings, there would be a loss of 240,000 shillings. Find the price value of her cupboard.

3. Find the selling price of a book if the buying price is:
   (a) 15,000 shillings and it is sold at a profit of 5%
   (b) 9,000 shillings and it is sold at a loss of 10%
   (c) 2,400 shillings and it is sold at a profit of 25%
   (d) 12,000 shillings and it is sold at a loss of 10%
   (e) 4,000 shillings and it is sold at a profit of 5%

4. Find the buying price of commodities if the selling price is:
   (a) 16,000 shillings and it is sold at a loss of 20%
   (b) 84,000 shillings and it is sold at a profit of 5%
   (c) 5,700,000 shillings and it is sold at a loss of 5%
   (d) 1,140,000 shillings and it is sold at a loss of 5%
   (e) 6,900,000 shillings and it is sold at a profit of 15%

5. John bought twenty chickens at 100,000 shillings and fifteen ducks at 126,500 shillings. He sold chickens at a profit of 25% and ducks at a profit of 15%. Find the total profit obtained.

6. Asia bought two hundred holiday cards at 3,000 shillings each. She sold the cards at 3,600 shillings each:

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(a) find the profit from the sales of all the cards

(b) what is the percentage profit?

7. Mshamu bought one hundred pawpaws from a farmer at 750 shillings each. He sold each pawpaw at 1,000 shillings. If all pawpaws were sold, what profit did Mshamu get?

8. Doto was charged 12% of his business capital as a penalty for failure to issue sales receipts to customers. If his capital was 8,800,000 shillings, what was the amount payed as penalty?

9. Jane bought an air ticket to Mwanza at a price of 350,000 shillings. If 14% of the air ticket price was paid as penalty for rescheduling her flight, how much money did she pay as penalty?

10. A businessman bought some commodities and allocated profit for each commodity as follows:

(a) one bag of beans; buying price of 240,000 shillings and a profit of 20%

(b) one bag of rice; buying price of 165,200 shillings and a profit of 18%

(c) one bag of maize; buying price of 77,000 shillings and a profit of 10%

(d) one bag of cowpeas; buying price 89,600 shillings and a profit of 12%

How much profit did he get after selling all commodities?

Invoice

An invoice is a list of items bought or sold with their prices or unit costs. Invoices are used to keep records of requirements, expenditures, sales and profits in a certain period. The symbol @

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which denotes "for each" is used when writing invoices. The symbol is usually placed between the quantity of an item and its unit price which means "price for each item". Therefore, the cost of the item is obtained by multiplying the quantity of items by the unit price.

Example 1

On 3rd December, 2025, Aneth was given shs 23,000 by her mother to buy school requirements as follows:

2 pairs of socks @ shs 1200

10 small exercise books @ shs 600

15 pens @ shs 250

5 pencils @ shs 100 and

4 erasers @ shs 150. Prepare an invoice of quantities and find her balance after expenditures.

Solution

Aneth's invoice on 3rd December, 2025 was as follows:

Items	**Cost (shs)**
2 pairs of socks @ shs 1,200	2,400
10 small exercise books @ shs 600	6,000
15 pens @ shs 250	3,750
5 pencils @ shs 100	500
4 erasers @ shs 150	600
Total	**13,250**

Aneth spent a total of 13,250 shillings. The amount left after expenditure is: $\text{shs } 23000 - \text{shs } 13250 = \text{shs } 9,750$

Therefore, her balance was 9,750 shillings.

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Example 2

After receiving a salary of 134,000 shillings on 26th April, 2025, Tumaini bought the following items:

6 kg of sugar @ shs 2,400

3 litres of cooking oil @ shs 1,200

10 kg of maize flour @ shs 900

7 kg of rice @ shs 1,600

Tumaini deposited the remaining amount of money in his bank account. Prepare Tumaini's invoice and then find the amount deposited in the bank account.

Solution

Tumaini's invoice on 26th April, 2025 was as follows:

Items	**Cost (shs)**
6 kg of sugar @ shs 2,400	14,400
3 litres of cooking oil @ shs 1,200	3,600
10 kg of maize flour @ shs 900	9,000
7 kg of rice @ shs 1,600	11,200
Total	**38,200**

Tumaini spent a total of 38,200 shillings

Amount received = shs 134,000

Expenditures = shs 38,200

The amount deposited in the bank

= amount received − expenditures

= shs 134000 − shs 38200

= shs 95,800

Therefore, the amount of money deposited in the bank account was shs 95,800.

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Exercise 2

1. Rose was given 23,000 shillings on 18th March, 2025. She bought 2 litres of milk @ shs 1,500
   2 kilograms of beans @ shs 2,500 and
   2 trays of eggs @ shs 7,500. Write an invoice for these items.

2. Atupele received a salary of 129,569 shillings on 23rd April, 2025 and bought the following items:
   3 pairs of trousers @ shs 15,000
   3 shirts @ shs 12,400
   1 belt @ shs 7,000 and
   2 bedsheets @ shs 3,000. Write an invoice of these items.

3. Subira bought the following items on 4th September, 2023:
   2 pairs of khanga @ shs 7,500
   1 pair of shoes @ shs 14,000
   3 pairs of socks @ sh 1,500
   3 tins of shoe polish @ shs 2,000 and
   2 airtime vouchers @ shs 1,000. If she had 100,000 shillings, what will her balance be?

4. On 20th May, 2025, John received 20,000 shillings from her sister for purchasing the following items:
   2 shoe brushes @ shs 400
   3 pieces of soap bar @ shs 250
   10 airtime vouchers @ shs 500
   10 bottles of soda @ shs 500
   5 kilograms of maize flour @ shs 1,000
   (a) write an invoice of these items
   (b) find the amount of money he remained with.

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5. Hamisi had 25,000 shillings on 1st November, 2024. He used some amount to buy the following items:
four cans of tomatoes at shs 600 each
3 pairs of slippers at shs 1,300 each
10 kilograms of wheat flour at shs 1,200 per kilogram
5 kilograms of cowpeas at shs 600 per kilogram and
2 kilograms of groundnuts at shs 800 per kilogram:
(a) prepare an invoice for these items
(b) find the amount of money Hamisi remained with.

6. On 14th July, 2025 the treasurer of a wedding committee collected 8,000,000 shillings for Juma's wedding ceremony. The budget for the ceremony was as follows:
drinks for 200 people @ shs 8,000
food for 200 people @ shs 10,000
reception hall shs 1,500,000
3 cars for transport @ shs 150,000:
(a) write an invoice for these items
(b) find the amount of money that remained after the ceremony.

7. On 20th May, 2025 a father bought the following items:
5 kilograms of sugar @ shs 2,500
10 litres of cooking oil @ shs 3,500
20 kilograms of rice @ shs 2,000 and
6 packets of salt @ shs 500:
(a) write an invoice of these items
(b) if the father had one hundred thousand shillings, how much money was he left with?

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8. Write an invoice of the following purchases done by the head teacher of a certain primary school on 7th January, 2025:

25 seedlings of flowers @ shs 700
12 counter books @ shs 3,000
12 rulers @ shs 250
2 thermos flasks @ shs 18,000 and
3 office curtains @ shs 8,000. How much money was paid for the purchases?

9. On 12th May, 2025 the sales of a fruits vendor were as follows:

55 oranges @ shs 200
10 watermelons @ shs 3,000
22 pawpaws @ shs 1,500
ripe bananas at shs 3,000
70 avocados @ shs 500 and
31 pineapples @ shs 1,000. How much money was collected by the fruits vendor?

10. On 14th May, 2025 a mother had 25,000 shillings. She used some of her money to buy the following items:

3 bunches of green vegetables @ shs 500
2 kilograms of meat @ shs 7,500
$\frac{1}{2}$ litre of cooking oil @ shs 3,500 and
$3\frac{1}{2}$ kilograms of maize flour @ shs 600:

(a) write an invoice for these items
(b) how much money did she remain with?

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Activity **Financial mathematics through online lessons**

Descriptions

Use various programs such as Khan Academy to find video lessons and practical exercises to deepen understanding on financial mathematics.

Summary

1. Profit is an excess amount of money obtained after selling goods.
2. Profit is the difference between the selling price and the buying price.
3. Loss is the opposite of profit.
4. An invoice is a list of items sold or bought and their prices.
5. The symbol @ is used when writing invoices to represent the words “for each”.

Revision exercise

1. Find the selling price of certain commodities if the buying price was:
   (a) 75,000 shillings and sold at a proft of 15%
   (b) 36,000 shillings and sold at a profit of 20%
   (c) 2,400,000 shillings and sold at a profit of 24%.

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2. Find the buying price of a commodity if the selling price was:
   (a) 32,000 shillings and sold at a loss of 20%
   (b) 840,000 shillings and sold at a profit of 12%.

3. Tumaini purchased a car at 7,250,000 shillings and sold it at 9,820,000 shillings:
   (a) how much profit did he get?
   (b) what was the percentage of the profit?

4. Anna was given 30,000 shillings on 17th April, 2025. She bought 4 litres of milk @ 2,000 shillings
   3 kilograms of beans @ 2,200 shillings and
   2 kilograms of rice @ 1,900 shillings.
   Write an invoice of these items.

5. Issa received a salary of 450,000 shillings on 25th May, 2025 and bought the following items:
   3 shirts @ shs 20,000
   2 pairs of trousers @ shs 32,400
   1 hat @ shs 4,500 and
   1 belt @ shs 3,000. Write an invoice of these items.

6. On 9th January, 2025, Thomas bought the following items:
   1 pair of shoes @ shs 14,000
   3 pairs of khanga @ shs 12,000
   3 pairs of socks @ shs 1,500
   2 tins of shoe polish @ shs 2,000 and
   2 airtime vouchers @ shs 1,000. If he had 90,000 shillings, how much money did he remain with?

7. On 14th May, 2025 Kabibi was given 20,000 shillings by her sister to buy the following requirements:
   2 bathing soaps @ shs 800
   3 washing soaps @ shs 250

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3 airtime vouchers @ shs 1,000
1 tube of toothpaste @ shs 1 500
(a) write an invoice of these items
(b) find the remaining amount of money.

8. On 22nd December, 2025 Mwajuma bought the following items:
3 kilograms of flour @ shs 1,200
5 litres of cooking oil @ shs 1,500
10 kilograms of sugar @ shs 2,000 and
3 packets of salt @ shs 500.
(a) write an invoice of these items
(b) if Mwajuma had 50,000 shillings, how much money did she remain with?

9. Write an invoice of the following purchases done by a farmer on 2nd September, 2025:
20 kg seeds of maize @ shs 2,000
12 kg seeds of beans @ shs 2,000
12 kg seeds of pumpkin @ shs 500
2 pairs of gumboots @ shs 12,000 and
3 hoes @ shs 8,000. How much money was paid for purchases?

10. On 2nd January, 2025, an entrepreneur sold a variety of fruits as follows:
50 mangoes @ shs 500
10 watermelons @ shs 2,000
20 pineapples @ shs 1,000
3 bunches of bananas @ shs 15,000
70 avocados @ shs 800 and
30 oranges @ shs 200. How much money did the entrepreneur get?

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